



Top 75 CAT Table with Missing values Questions With Video Solutions

All rights reserved. No part of this publication may be reproduced, distributed, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, or stored in any retrieval system of any nature without the permission of cracku.in, application for which shall be made to support@cracku.in

Questions

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.

The Dean's office recently scanned student results into the central computer system. When their character reading software cannot read something, it leaves the space blank. The scanner output read as follows:

Name	Finance	Marketing	Statistics	Strategy	Operations	GPA
Aparna		B	F			1.4
Bikas	D	D	F	F		
Chandra		D	A	F	F	2.4
Deepak	A	B		D	D	3.2
Fazal	D	F		B	D	2.4
Gowri	C	C	A		B	3.8
Hari		B	A		D	2.8
Ismet			B		A	
Jagdeep	A	A	B		C	3.8
Kunal	F		A	F	F	1.8
Leena	B	A		B	F	3.2
Manab			A	B	B	
Nisha	A	D	B	A	F	3.6
Osman	C		B	B	A	4.6
Preeti	F	D		D		3.2
Rahul	A	C	A		F	4.2
Sameer		C	F	B		
Tara	B					2.4
Utkarsh			F	C	A	3.0
Vipul	A		C	C	F	2.4

In the grading system, A, B, C, D, and F grades fetch 6, 4, 3, 2, and 0 grade points respectively. The Grade Point Average (GPA) is the arithmetic mean of the grade points obtained in the five subjects. For example Nisha's GPA is $(6 + 2 + 4 + 6 + 0) / 5 = 3.6$. Some additional facts are also known about the students' grades. These are

- Vipul obtained the same grade in Marketing as Aparna obtained in Finance and Strategy.
- Fazal obtained the same grade in Strategy as Utkarsh did in Marketing.
- Tara received the same grade in exactly three courses.

Question 1

What grade did Preeti obtain in Statistics?

- A A
- B B
- C C
- D D

Directions for the following four questions: Answer the questions on the basis of the information given below.

The Dean's office recently scanned student results into the central computer system. When their character reading software cannot read something, it leaves the space blank. The scanner output read as follows:

Name	Finance	Marketing	Statistics	Strategy	Operations	GPA
Aparna		B	F			1.4
Bikas	D	D	F	F		
Chandra		D	A	F	F	2.4
Deepak	A	B		D	D	3.2
Fazal	D	F		B	D	2.4
Gowri	C	C	A		B	3.8
Hari		B	A		D	2.8
Ismet			B		A	
Jagdeep	A	A	B		C	3.8
Kunal	F		A	F	F	1.8
Leena	B	A		B	F	3.2
Manab			A	B	B	
Nisha	A	D	B	A	F	3.6
Osman	C		B	B	A	4.6
Preeti	F	D		D		3.2
Rahul	A	C	A		F	4.2
Sameer		C	F	B		
Tara	B					2.4
Utkarsh			F	C	A	3.0
Vipul	A		C	C	F	2.4

In the grading system, A, B, C, D, and F grades fetch 6, 4, 3, 2, and 0 grade points respectively. The Grade Point Average (GPA) is the arithmetic mean of the grade points obtained in the five subjects. For example Nisha's GPA is $(6 + 2 + 4 + 6 + 0) / 5 = 3.6$. Some additional facts are also known about the students' grades. These are

- Vipul obtained the same grade in Marketing as Aparna obtained in Finance and Strategy.
- Fazal obtained the same grade in Strategy as Utkarsh did in Marketing.
- Tara received the same grade in exactly three courses.

cracku

A → 6
B → 4
C → 3
D → 2
F → 0

Instructions

Funky Pizzeria was required to supply pizzas to three different parties. The total number of pizzas it had to deliver was 800, 70% of which were to be delivered to Party 3 and the rest equally divided between Party 1 and Party 2.

Pizzas could be of Thin Crust (T) or Deep Dish (D) variety and come in either Normal Cheese (NC) or Extra Cheese (EC) versions. Hence, there are four types of pizzas: T-NC, T-EC, D-NC and D-EC. Partial information about proportions of T and NC pizzas ordered by the three parties is given below:

	Thin crust (T)	Normal cheese (NC)
Party 1	0.6	
Party 2	0.55	0.3
Party 3		0.65
Total	0.375	0.52

Question 2

For Party 2, if 50% of the Normal Cheese pizzas were of Thin Crust variety, what was the difference between the numbers of T-EC and D-EC pizzas to be delivered to Party 2?

- A 18
- B 12
- C 30
- D 24

Cracku

Party 1 and Party 2

Pizzas could be of Thin Crust (T) or Deep Dish (D) variety and come in either Normal Cheese (NC) or Extra Cheese (EC) versions. Hence, there are four types of pizzas: T-NC, T-EC, D-NC and D-EC. Partial information about proportions of T and NC pizzas ordered by the three parties is given below:

	Thin crust (T)	Normal cheese (NC)
Party 1	0.6	
Party 2	0.55	0.3
Party 3		0.65
Total	0.375	0.52

Handwritten notes:

15/100 = 15%
15/100 = 15%
15/100 = 15%

T-EC, D-EC, T-D, T-NC, D-NC, NC-EC

Party 1
Party 2
Party 3

Instructions

A company administers a written test comprising of three sections of 20 marks each - Data Interpretation (DI), Written English (WE) and General Awareness (GA), for recruitment. A composite score for a candidate (out of 80) is calculated by doubling her marks in DI and adding it to the sum of her marks in the other two sections. Candidates who score less than 70% marks in two or more sections are disqualified. From among the rest, the four with the highest composite scores are recruited. If four or less candidates qualify, all who qualify are recruited.

Ten candidates appeared for the written test. Their marks in the test are given in the table below. Some marks in the table are missing, but the following facts are known:

1. No two candidates had the same composite score.
2. Ajay was the unique highest scorer in WE.
3. Among the four recruited, Geeta had the lowest composite score.
4. Indu was recruited.
5. Danish, Harini, and Indu had scored the same marks the in GA.
6. Indu and Jatin both scored 100% in exactly one section and Jatin's composite score was 10 more than Indu's.

Candidate	Marks out of 20		
	DI	WE	GA
Ajay	8		16
Bala		9	11
Chetna	19	4	12
Danish	8	15	
Ester	12	18	16
Falak	15	7	10
Geeta	14		6
Harini	5		
Indu		8	
Jatin		16	14

Question 3

If all the candidates scored different marks in WE then what is the maximum marks that Harini could have scored in WE?

1. No two candidates had the same composite score.

2. Ajay was the unique highest scorer in WE.

3. Among the four recruited, Geeta had the lowest composite score.

4. Indu was recruited.

5. Danish, Harini, and Indu had scored the same marks the in GA.

6. Indu and Jatin both scored 100% in exactly one section and Jatin's composite score was 10 more than Indu's.

Candidate	DI	WE	GA
Ajay	8		16
Bala		9	11
Chetna	19	4	12
Danish	8	15	
Ester	12	18	16
Falak	15	7	10
Geeta	14		6
Harini	5		
Indu		8	
Jatin		16	14

Handwritten notes:

CS = 2x DI (20) + WE (20) + GA (20) = 80

Cracku logo



VIDEO SOLUTION

CAT 2025 MOCKS & SECTIONALS
WITH VIDEO SOLUTIONS & ANALYSIS

₹ 2999

ENROLL NOW

Instructions

The year is 2089. Beijing, London, New York, and Paris are in contention to host the 2096 Olympics. The eventual winner is determined through several rounds of voting by members of the IOC with each member representing a different city. All the four cities in contention are also represented in IOC.

Round	Total Votes Cast	Maximum Votes Cast		Eliminated	
		City	No. of Votes	City	No. of Votes
1		London	30	New York	12
2	83	Paris	32	Beijing	21
3	75				

- In any round of voting, the city receiving the lowest number of votes in that round gets eliminated. The survivor after the last round of voting gets to host the event.

- A member is allowed to cast votes for at most two different cities in all rounds of voting combined. (Hence, a member becomes ineligible to cast a vote in a given round if both the cities (s)he voted for in earlier rounds are out of contention in that round of voting.)
- A member is also ineligible to cast a vote in a round if the city (s)he represents is in contention in that round of voting.
- As long as the member is eligible, (s)he must vote and vote for only one candidate city in any round of voting. The following incomplete table shows the information on cities that received the maximum and minimum votes in different rounds, the number of votes cast in their favour, and the total votes that were cast in those rounds. It is also known that:
 - All those who voted for London and Paris in round 1, continued to vote for the same cities in subsequent rounds as long as these cities were in contention. 75% of those who voted for Beijing in round 1, voted for Beijing in round 2 as well.
 - Those who voted for New York in round 1, voted either for Beijing or Paris in round 2.
 - The difference in votes cast for the two contending cities in the last round was 1.
 - 50% of those who voted for Beijing in round 1, voted for Paris in round 3.

Question 4

Which of the following statements must be true?

- IOC member from New York must have voted for Paris in round 2.
- IOC member from Beijing voted for London in round 3.

- Only a
- Only b
- Both a and b
- Neither a nor b

cracku

Read the following information carefully and answer the questions which follow.

The year is 2089. Beijing, London, New York, and Paris are in contention to host the 2096 Olympics. The eventual winner is determined through several rounds of voting by members of the IOC with each member representing a different city. All the four cities in contention are also represented in IOC.

In any round of voting, the city receiving the lowest number of votes in that round gets eliminated. The survivor after the last round of voting gets to host the event.

A member is allowed to cast votes for at most two different cities in all rounds of voting combined. (Hence, a member becomes ineligible to cast a vote in a given round if both the cities (s)he voted for in earlier rounds are out of contention in that round of voting.) A member is also ineligible to cast a vote in a round if the city (s)he represents is in contention in that round of voting. As long as the member is eligible, (s)he must vote and vote for only one candidate city in any round of voting. The following incomplete table shows the information on cities that received the maximum and minimum votes in different rounds, the number of votes cast in their favour, and the total votes that were cast in those rounds.

Round	Total Votes	Maximum Votes Cast		Eliminated	
		City	No. of Votes	City	No. Of Votes
1		London	30	New York	12
2	83	Paris	32	Beijing	21
3	75				

VIDEO SOLUTION

Instructions

Six players - Tanzi, Umeza, Wangdu, Xyla, Yonita and Zeneca competed in an archery tournament. The tournament had three compulsory rounds, Rounds 1 to 3. In each round every player shot an arrow at a target. Hitting the centre of the target (called bull's eye) fetched the highest score of 5. The only other possible scores that a player could achieve were 4, 3, 2 and 1. Every bull's eye score in the first three rounds gave a player one additional chance to shoot in the bonus rounds, Rounds 4 to 6. The possible scores in Rounds 4 to 6 were identical to the first three.

A player's total score in the tournament was the sum of his/her scores in all rounds played by him/her. The table below presents partial information on points scored by the players after completion of the tournament. In the table, NP means that the player did not participate in that round, while a hyphen means that the player participated in that round and the score information is missing.

	Round-1	Round-2	Round-3	Round-4	Round-5	Round-6
Tanzi	-	4	-	5	NP	NP
Umeza	-	-	-	1	2	NP
Wangdu	-	4	-	NP	NP	NP
Xyla	-	-	-	1	5	-
Yonita	-	-	3	5	NP	NP
Zeneca	-	-	-	5	5	NP

The following facts are also known.

1. Tanzi, Umeza and Yonita had the same total score.
2. Total scores for all players, except one, were in multiples of three.
3. The highest total score was one more than double of the lowest total score.
4. The number of players hitting bull's eye in Round 2 was double of that in Round 3.
5. Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

Question 5

What was the highest total score?

- A 25
- B 21
- C 24
- D 23

The following facts are also known.

1. Tanzi, Umeza and Yonita had the same total score.
2. Total scores for all players, except one, were in multiples of three.
3. The highest total score was one more than double of the lowest total score.
4. The number of players hitting bull's eye in Round 2 was double of that in Round 3.
5. Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

VIDEO SOLUTION

Question 6

Which of the following statements is true?

- A Xyla's score was 23.
- B Zeneca's score was 23.
- C Zeneca was the highest scorer.
- D Xyla was the highest scorer.

As player's total score in the tournament was the sum of his/her scores in all rounds played by him/her. The table below presents partial information on points scored by the players after completion of the tournament. In the table, NP means that the player did not participate in that round, while a hyphen means that the player participated in that round and the score information is missing.

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	-	4	-	5	NP	NP	-
Umeza	-	-	-	1	2	NP	-
Wangde	-	4	-	NP	NP	NP	-
Xyla	-	-	-	1	-	-	-
Yonita	-	-	8	5	NP	NP	-
Zeneca	-	-	-	9	-	-	-

The following facts are also known:

1. Tanzi, Umeza and Yonita had the same total score.
2. Total scores for all players, except one, were in multiples of three.
3. The highest total score was one more than double of the lowest total score.
4. The number of players hitting bull's eye in Round 2 was double of that in Round 3.
5. Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

VIDEO SOLUTION

CAT Previous Papers PDF

Question 7

What was Tanzi's score in Round 3?

- A 4
- B 5
- C 3
- D 1

As player's total score in the tournament was the sum of his/her scores in all rounds played by him/her. The table below presents partial information on points scored by the players after completion of the tournament. In the table, NP means that the player did not participate in that round, while a hyphen means that the player participated in that round and the score information is missing.

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	-	4	-	5	NP	NP	-
Umeza	-	-	-	1	2	NP	-
Wangde	-	4	-	NP	NP	NP	-
Xyla	-	-	-	1	-	-	-
Yonita	-	-	8	5	NP	NP	-
Zeneca	-	-	-	9	-	-	-

The following facts are also known:

1. Tanzi, Umeza and Yonita had the same total score.
2. Total scores for all players, except one, were in multiples of three.
3. The highest total score was one more than double of the lowest total score.
4. The number of players hitting bull's eye in Round 2 was double of that in Round 3.
5. Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

VIDEO SOLUTION

Instructions

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of these people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran		1	
Dhinesh			2
Ezhil			1
Fani	1		
Gowtham		3	2

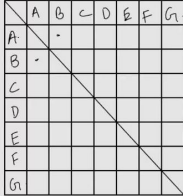
The following additional facts are also known.

1. Amudha, Bharatan, and Chandran are mutual strangers.
2. Amudha, Dhinesh, and Fani are Ezil's friends.
3. Chandran and Gowtham are friends.
4. Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
5. Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

Question 8

How many people are either a friend or a friend-of-a-friend of Ezhil?



Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of these people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran		1	
Dhinesh			2
Ezhil			1
Fani	1		
Gowtham		3	2

The following additional facts are also known.

1. Amudha, Bharatan, and Chandran are mutual strangers.
2. Amudha, Dhinesh, and Fani are Ezil's friends.
3. Chandran and Gowtham are friends.
4. Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
5. Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.



VIDEO SOLUTION

Instructions

Funky Pizzeria was required to supply pizzas to three different parties. The total number of pizzas it had to deliver was 800, 70% of which were to be delivered to Party 3 and the rest equally divided between Party 1 and Party 2.

Pizzas could be of Thin Crust (T) or Deep Dish (D) variety and come in either Normal Cheese (NC) or Extra Cheese (EC) versions. Hence, there are four types of pizzas: T-NC, T-EC, D-NC and D-EC. Partial information about proportions of T and NC pizzas ordered by the three parties is given below:

	Thin crust (T)	Normal cheese (NC)
Party 1	0.6	
Party 2	0.55	0.3
Party 3		0.65
Total	0.375	0.52

Question 9

Suppose that a T-NC pizza cost as much as a D-NC pizza, but $\frac{3}{5}$ th of the price of a D-EC pizza. A D-EC pizza costs Rs. 50 more than a T-EC pizza, and the latter costs Rs. 500.

If 25% of the Normal Cheese pizzas delivered to Party 1 were of Deep Dish variety, what was the total bill for Party 1?

- D** Rs. 45240

Cheese Pizzas and Party 2

Pizzas could be of Thin Crust (T) or Deep Dish (D) variety and come in either Normal Cheese (NC) or Extra Cheese (EC) versions. Hence, there are four types of pizzas: T-NC, T-EC, D-NC and D-EC. Partial information about proportions of T and NC pizzas ordered by the three parties is given below:

	Thin crust (T)	Normal cheese (NC)
Party 1	0.8	
Party 2	0.55	0.1
Party 3		0.65
Total	0.375	0.52

Handwritten notes on the left side of the slide:

- 11
- $\downarrow$
- $15\% = 120$
- $10\% = 560$
- T-EC, D-EC, T-NC, D-NC
- T-NC, D-NC, NC-EC

A large play button icon is centered on the slide.

At the bottom right, a person is visible sitting at a desk, looking at a laptop screen.



Instructions

A company administers a written test comprising of three sections of 20 marks each - Data Interpretation (DI), Written English (WE) and General Awareness (GA), for recruitment. A composite score for a candidate (out of 80) is calculated by doubling her marks in DI and adding it to the sum of her marks in the other two sections. Candidates who score less than 70% marks in two or more sections are disqualified. From among the rest, the four with the highest composite scores are recruited. If four or less candidates qualify, all who qualify are recruited.

Ten candidates appeared for the written test. Their marks in the test are given in the table below. Some marks in the table are missing, but the following facts are known:

1. No two candidates had the same composite score.
2. Ajay was the unique highest scorer in WE.
3. Among the four recruited, Geeta had the lowest composite score.
4. Indu was recruited.
5. Danish, Harini, and Indu had scored the same marks in GA.
6. Indu and Jatin both scored 100% in exactly one section and Jatin's composite score was 10 more than Indu's.

Candidate	Marks out of 20		
	DI	WE	GA
Ajay	8		16
Bala		9	11
Chetna	19	4	12
Danish	8	15	
Ester	12	18	16
Falak	15	7	10
Geeta	14		6
Harini	5		
Indu		8	
Jatin		16	14

Question 10

Which of the following statements MUST be true?

1. Jatin's composite score was more than that of Danish.
2. Indu scored less than Chetna in DI.
3. Jatin scored more than Indu in GA.

- A** Both 2 and 3
- B** Only 1
- C** Only 2
- D** Both 1 and 2

1. No two candidates had the same composite score.
2. Ajay was the unique highest scorer in WE.
3. Among the four recruited, Geeta had the lowest composite score.
4. Indu was recruited.
5. Danish, Harini, and Indu had scored the same marks in GA.
6. Indu and Jatin both scored 100% in exactly one section and Jatin's composite score was 10 more than Indu's.

$$\begin{aligned}
 \text{CS} &= 2 \times \text{DI} \quad (20) \\
 &+ \text{WE} \quad (20) \\
 &+ \text{GA} \quad (20) \\
 &= 80
 \end{aligned}$$

Cracku

Candidate	DI	WE	GA
Ajay	8		16
Bala		9	11
Chetna	19	4	12
Danish	8	15	
Ester	12	18	16
Falak	15	7	10
Geeta	14		6
Harini	5		
Indu		8	
Jatin		16	14

VIDEO SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.

The Dean's office recently scanned student results into the central computer system. When their character reading software cannot read something, it leaves the space blank. The scanner output read as follows:

Name	Finance	Marketing	Statistics	Strategy	Operations	GPA
Aparna		B	F			1.4
Bikas	D	D	F	F		
Chandra		D	A	F	F	2.4
Deepak	A	B		D	D	3.2
Fazal	D	F		B	D	2.4
Gowri	C	C	A		B	3.8
Hari		B	A		D	2.8
Ismet			B		A	
Jagdeep	A	A	B		C	3.8
Kunal	F		A	F	F	1.8
Leena	B	A		B	F	3.2
Manab			A	B	B	
Nisha	A	D	B	A	F	3.6
Osman	C		B	B	A	4.6
Preeti	F	D		D		3.2
Rahul	A	C	A		F	4.2
Sameer		C	F	B		
Tara	B					2.4
Utkarsh			F	C	A	3.0
Vipul	A		C	C	F	2.4

In the grading system, A, B, C, D, and F grades fetch 6, 4, 3, 2, and 0 grade points respectively. The Grade Point Average (GPA) is the arithmetic mean of the grade points obtained in the five subjects. For example Nisha's GPA is $(6 + 2 + 4 + 6 + 0) / 5 = 3.6$. Some additional facts are also known about the students' grades. These are

- (i) Vipul obtained the same grade in Marketing as Aparna obtained in Finance and Strategy.
- (ii) Fazal obtained the same grade in Strategy as Utkarsh did in Marketing.
- (iii) Tara received the same grade in exactly three courses.

Question 11

In Strategy, Gowri's grade point was higher than that obtained by

- A Fazal
- B Hari
- C Nisha
- D Rahul

Directions for the following four questions: Answer the questions on the basis of the information given below.

The Dean's office recently scanned student results into the central computer system. When their character reading software cannot read something, it leaves the space blank. The scanner output read as follows:

Name	Finance	Marketing	Statistics	Strategy	Operations	GPA
Aparna		B	F			1.4
Bikas	D	D	F	F		
Chandra		D	A	F	F	2.4
Deepak	A	B		D	D	3.2
Fazal	D	F		B	D	2.4
Gowri	C	C	A		B	3.8
Hari		B	A		D	2.8
Ismet			B		A	
Jagdeep	A	A	B		C	3.8
Kunal	F		A	F	F	1.8
Leena	B	A		B	F	3.2
Manab			A	B	B	
Nisha	A	D	B	A	F	3.6
Osman	C		B	B	A	4.6
Preeti	F	D		D		3.2
Rahul	A	C	A		F	4.2
Sameer		C	F	B		
Tara	B					2.4
Utkarsh			F	C	A	3.0
Vipul	A		C	C	F	2.4

In the grading system, A, B, C, D, and F grades fetch 6, 4, 3, 2, and 0 grade points respectively. The Grade Point Average (GPA) is the arithmetic mean of the grade points obtained in the five subjects. For example Nisha's GPA is $(6 + 2 + 4 + 6 + 0) / 5 = 3.6$. Some additional facts are also known about the students' grades. These are

(i) Vipul obtained the same grade in Marketing as Aparna obtained in Finance and Strategy.

cracku

A → 6
B → 4
C → 3
D → 2
F → 0

VIDEO SOLUTION

Question 12

What grade did Utkarsh obtain in Finance?

- A B

D F

 VIDEO SOLUTION

Instructions

	Total Earning				Total Days			
Emp No.	Complex	Medium	Simple	Total	Complex	Medium	Simple	Total
2001147	82.98		636.53	719.51	3	0	23	26
2001148	51.53		461.73	513.26	3.3	1.67	16	21
2001149	171.10		79.10	250.81	5.5	4	8.5	18
2001150	100.47		497.47	597.95	6	4.67	7.33	18
2001151	594.43	159.64		754.06	9.67	13.33	0	23
2001156	83.70			89.70	8	0	1	9
2001158	472.31	109.73		582.04	1.39	9.61	0	11
2001164	402.25	735.22	213.67	1351.14	5.27	12.07	0.67	18
2001170	576.57			576.57	21	0	0	21
2001171	286.47	6.10		292.57	8.38	4.25	0.38	13
2001172	512.10	117.46		629.56	10	8.5	3.5	22
2001173	1303.88			1303.88	25.5	0	0.5	26
2001174	1017.94			1017.9	26	0	0	26
2001179	46.56	776.19		822.75	2	19	0	21
2001180	116.40	1262.79		1379.19	5	19	0	24

Among the employees who were engaged in complex and medium operations, the number of employees whose average earning per day in complex operations is more than average earning per day in medium operations is

C 5

Instructions

The year is 2089. Beijing, London, New York, and Paris are in contention to host the 2096 Olympics. The eventual winner is determined through several rounds of voting by members of the IOC with each member representing a different city. All the four cities in contention are also represented in IOC.

Round	Total Votes Cast	Maximum Votes Cast		Eliminated	
		City	No. of Votes	City	No. of Votes
1		London	30	New York	12
2	83	Paris	32	Beijing	21
3	75				

- In any round of voting, the city receiving the lowest number of votes in that round gets eliminated. The survivor after the last round of voting gets to host the event.
- A member is allowed to cast votes for at most two different cities in all rounds of voting combined. (Hence, a member becomes ineligible to cast a vote in a given round if both the cities (s)he voted for in earlier rounds are out of contention in that round of voting.)
- A member is also ineligible to cast a vote in a round if the city (s)he represents is in contention in that round of voting.
- As long as the member is eligible, (s)he must vote and vote for only one candidate city in any round of voting. The following incomplete table shows the information on cities that received the maximum and minimum votes in different rounds, the number of votes cast in their favour, and the total votes that were cast in those rounds. It is also known that:
- All those who voted for London and Paris in round 1, continued to vote for the same cities in subsequent rounds as long as these cities were in contention. 75% of those who voted for Beijing in round 1, voted for Beijing in round 2 as well.
- Those who voted for New York in round 1, voted either for Beijing or Paris in round 2.
- The difference in votes cast for the two contending cities in the last round was 1.
- 50% of those who voted for Beijing in round 1, voted for Paris in round 3.

Question 14

What percentage of members from among those who voted for New York in round 1, voted for Beijing in round 2?

- A 31.33
- B 50
- C 66.67
- D 75

Read the following information carefully and answer the questions which follow.

The year is 2089. Beijing, London, New York, and Paris are in contention to host the 2096 Olympics. The eventual winner is determined through several rounds of voting by members of the IOC with each member representing a different city. All the four cities in contention are also represented in IOC.

In any round of voting, the city receiving the lowest number of votes in that round gets eliminated. The survivor after the last round of voting gets to host the event.

A member is allowed to vote for at most two different cities in all rounds of voting combined. (Hence, a member is ineligible to cast a vote in a given round if both the cities (s)he voted for are still in contention in that round of voting.) A member is also ineligible to cast a vote if the city (s)he represents is in contention in that round of voting. As long as a member is eligible, (s)he must vote and vote for only one candidate city in any round of voting. The following incomplete table shows the information on cities that received the maximum and minimum votes in different rounds, the number of votes cast in their favor, and the total votes that were cast in those rounds.

Round	Total votes	Maximum Votes Cast		Eliminated	
		City	No. of Votes	City	No. Of Votes
1		London	30	New York	12
2	83	Paris	32	Beijing	21
3	75				

VIDEO SOLUTION

Instructions

An agency entrusted to accredit colleges looks at four parameters: faculty quality (F), reputation (R), placement quality (P), and infrastructure (I). The four parameters are used to arrive at an overall score, which the agency uses to give an accreditation to the colleges. In each parameter, there are five possible letter grades given, each carrying certain points: A (50 points), B (40 points), C (30 points), D (20 points), and F (0 points). The overall score for a college is the weighted sum of the points scored in the four parameters. The weights of the parameters are 0.1, 0.2, 0.3 and 0.4 in some order, but the order is not disclosed. Accreditation is awarded based on the following scheme:

Range	Accreditation
Overall score ≥ 45	AAA
$35 \leq \text{Overall score} < 45$	BAA
$25 \leq \text{Overall score} < 35$	BBA
$15 \leq \text{Overall score} < 25$	BBB
Overall score < 15	Junk

Eight colleges apply for accreditation, and receive the following grades in the four parameters (F, R, P, and I):

	F	R	P	I
A-one	A	A	A	B
Best Ed	B	C	D	D
Cosmopolitan	B	D	D	C
Dominance	D	D	B	C
Education Aid	A	A	B	A
Fancy	A	A	B	B
Global	C	F	D	D
High Q	C	D	D	B

It is further known that in terms of overall scores:

1. High Q is better than Best Ed;
2. Best Ed is better than Cosmopolitan; and
3. Education Aid is better than A-one.

Question 15

What is the highest overall score among the eight colleges ?

cracku CAT 2025 June

Eight colleges apply for accreditation, and receive the following grades in the four parameters (F, R, P, and I):

	F	R	P	I
A-one	A	A	A	B
Best Ed	B	C	D	D
Cosmopolitan	B	D	D	C
Dominance	D	D	B	C
Education Aid	A	A	B	A
Fancy	A	A	B	B
Global	C	F	D	B
High-Q	C	D	D	B

It is further known that in terms of overall scores:

1. High-Q is better than Best Ed;
2. Best Ed is better than Cosmopolitan; and
3. Education Aid is better than A-one.

Handwritten notes on the right:

A-50
B-40
C-30
D-20
E-0

0.1 → F, R, P, I
0.2 →
0.3 →
0.4 →



VIDEO SOLUTION

FREE DOWNLOAD **CAT 2025 SYLLABUS PDF** PDF

Question 16

How many colleges receive the accreditation of AAA?

cracku CAT 2025 June

Eight colleges apply for accreditation, and receive the following grades in the four parameters (F, R, P, and I):

	F	R	P	I
A-one	A	A	A	B
Best Ed	B	C	D	D
Cosmopolitan	B	D	D	C
Dominance	D	D	B	C
Education Aid	A	A	B	A
Fancy	A	A	B	B
Global	C	F	D	B
High-Q	C	D	D	B

It is further known that in terms of overall scores:

1. High-Q is better than Best Ed;
2. Best Ed is better than Cosmopolitan; and
3. Education Aid is better than A-one.

Handwritten notes on the right:

A-50
B-40
C-30
D-20
E-0

0.1 → F, R, P, I
0.2 →
0.3 →
0.4 →



VIDEO SOLUTION

Instructions

The following table gives details regarding the total earnings of 15 employees and the number of days they have worked on complex, medium and simple operation in the month of June 2002. Even though the employees might have worked on an operation, they would be eligible for earnings only if they have minimum level of efficiency.

Emp No.	Total Earning				Total Days			
	Complex	Medium	Simple	Total	Complex	Medium	Simple	Total
2001147	82.98		636.53	719.51	3	0	23	26
2001148	51.53		461.73	513.26	3.3	1.67	16	21
2001149	171.10		79.10	250.81	5.5	4	8.5	18
2001150	100.47		497.47	597.95	6	4.67	7.33	18
2001151	594.43	159.64		754.06	9.67	13.33	0	23
2001156	83.70			89.70	8	0	1	9
2001158	472.31	109.73		582.04	1.39	9.61	0	11
2001164	402.25	735.22	213.67	1351.14	5.27	12.07	0.67	18
2001170	576.57			576.57	21	0	0	21
2001171	286.47	6.10		292.57	8.38	4.25	0.38	13
2001172	512.10	117.46		629.56	10	8.5	3.5	22
2001173	1303.88			1303.88	25.5	0	0.5	26
2001174	1017.94			1017.9	26	0	0	26
2001179	46.56	776.19		822.75	2	19	0	21
2001180	116.40	1262.79		1379.19	5	19	0	24

Question 17

The employee number of the person who has earned the maximum earnings per day in medium operation is

- A 2001180
- B 2001164
- C 2001172
- D 2001179

[VIEW SOLUTION](#)

Question 18

The number of employees who have earned more than Rs. 600 and having more than 80% attendance (there are 25 regular working days in June 2002; some might be coming on overtime too) is

- A 4
- B 5
- C 6
- D 7

[VIEW SOLUTION](#)

[CAT Formulas PDF \[Download Now\]](#)

Question 19

The number of employees who have earned more than Rs. 50 per day in complex operations is

- A 4
- B 3
- C 5
- D 6

Instructions

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of these people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran		1	
Dhinesh			2
Ezhil			1
Fani	1		
Gowtham		3	2

The following additional facts are also known.

1. Amudha, Bharatan, and Chandran are mutual strangers.
2. Amudha, Dhinesh, and Fani are Ezhil's friends.
3. Chandran and Gowtham are friends.
4. Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
5. Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

Question 20

How many friends does Ezhil have?



	A	B	C	D	E	F	G
A							
B							
C							
D							
E							
F							
G							

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of these people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran		1	
Dhinesh			2
Ezhil			1
Fani	1		
Gowtham		3	2

The following additional facts are also known.

1. Amudha, Bharatan, and Chandran are mutual strangers.
2. Amudha, Dhinesh, and Fani are Ezhil's friends.
3. Chandran and Gowtham are friends.
4. Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
5. Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

Instructions

Odsville has five firms - Alfloo, Bzygoo, Czechy, Drjbna and Elavalaki. Each of these firms was founded in some year and also closed down a few years later.

Each firm raised Rs. 1 crore in its first and last year of existence. The amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down. No firm raised the same amount of money in two consecutive years. Each annual increase and decrease was either by Rs. 1 crore or by Rs. 2 crores. The table below provides partial information about the five firms.

Firm	First year of existence	Last year of existence	Total amount raised (Rs. Crores)
Alfloo	2009	2016	21
Bzygoo	2012	2015	
Czechy	2013		9
Drybna	2011	2015	10
Elavalaki	2010		13

Question 21

What is the largest possible total amount of money (in Rs. crores) that could have been raised in 2013?

cracku

Odville has five firms - Alfloo, Bzygoo, Czechy, Drybna and Elavalaki. Each of these firms was founded in some year and also closed down a few years later.

Each firm raised Rs. 1 crore in its first and last year of existence. The amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down. No firm raised the same amount of money in two consecutive years. The annual increase and decrease was either by Rs. 1 crore or by Rs. 2 crores.

The table below provides partial information about the five firms.

Firm	First year of existence	Last year of existence	Total amount raised (Rs. Crores)
Alfloo	2009	2016	21
Bzygoo	2012	2015	
Czechy	2013		9
Drybna	2011	2015	10
Elavalaki	2010		13

VIDEO SOLUTION



CAT FORMULAS BOOK

BY 100%ILER



Instructions

Six players - Tanzi, Umeza, Wangdu, Xyla, Yonita and Zeneca competed in an archery tournament. The tournament had three compulsory rounds, Rounds 1 to 3. In each round every player shot an arrow at a target. Hitting the centre of the target (called bull's eye) fetched the highest score of 5. The only other possible scores that a player could achieve were 4, 3, 2 and 1. Every bull's eye score in the first three rounds gave a player one additional chance to shoot in the bonus rounds, Rounds 4 to 6. The possible scores in Rounds 4 to 6 were identical to the first three.

A player's total score in the tournament was the sum of his/her scores in all rounds played by him/her. The table below presents partial information on points scored by the players after completion of the tournament. In the table, NP means that the player did not participate in that round, while a hyphen means that the player participated in that round and the score information is missing.

	Round-1	Round-2	Round-3	Round-4	Round-5	Round-6
Tanzi	-	4	-	5	NP	NP
Umeza	-	-	-	1	2	NP
Wangdu	-	4	-	NP	NP	NP
Xyla	-	-	-	1	5	-
Yonita	-	-	3	5	NP	NP
Zeneca	-	-	-	5	5	NP

The following facts are also known.

1. Tanzi, Umeza and Yonita had the same total score.
2. Total scores for all players, except one, were in multiples of three.

3. The highest total score was one more than double of the lowest total score.
 4. The number of players hitting bull's eye in Round 2 was double of that in Round 3.
 5. Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

Question 22

What was Zeneca's total score?

- A 21
 B 22
 C 23
 D 24

Cracku

A player's total score in a tournament is the sum of the scores in the rounds played by him/her. The table below presents partial information on points scored by the players after completion of the tournament. In the table, NP means that the player did not participate in that round, while a hyphen means that the player participated in that round and the score information is missing.

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	-	4	-	5	NP	NP	-
Uma	-	-	1	-	2	NP	-
Wangle	-	4	-	NP	NP	NP	-
Xia	-	-	-	1	-	-	-
Yonita	-	-	3	5	-	NP	-
Zeneca	-	-	-	5	-	NP	-

The following facts are also known.

1. Tanzi, Uma and Yonita had the same total score.
2. Total scores for all players, except one, were in multiples of three.
3. The highest total score was one more than double of the lowest total score.
4. The number of players hitting bull's eye in Round 2 was double of that in Round 3.
5. Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

VIDEO SOLUTION

Instructions

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers -

Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	1 and 4
Range of rating*	3	3	4	4	3

* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

- (a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.
 (b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

Question 23

What is the median of the ratings given by R3 to the five workers?

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers - Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	3 and 4
Range of rating*	3	3	4	4	3

* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

(a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.

(b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

Handwritten notes on the right show the calculation of the mean rating for R1: $1, 2, 3, 4, 5$ and $\text{Mean} = \frac{1+2+3+4+5}{5} = 3$. Below this, the word "Median" is written, and the numbers 1, 2, 3, 4, 5 are listed with the number 2 circled, indicating the median rating.

VIDEO SOLUTION

Instructions

In a coaching class, some students register online, and some others register offline. No student registers both online and offline; hence the total registration number is the sum of online and offline registrations. The following facts and table pertain to these registration numbers for the five months - January to May of 2023. The table shows the minimum, maximum, median registration numbers of these five months, separately for online, offline and total number of registrations. The following additional facts are known.

1. In every month, both online and offline registration numbers were multiples of 10.
2. In January, the number of offline registrations was twice that of online registrations.
3. In April, the number of online registrations was twice that of offline registrations.
4. The number of online registrations in March was the same as the number of offline registrations in February.
5. The number of online registrations was the largest in May.

	Minimum	Maximum	Median
Online	40	100	80
Offline	30	80	50
Total	110	130	120

Question 24

What best can be concluded about the number of offline registrations in February?

- A 80
- B 50 or 80
- C 30 or 50 or 80
- D 50

shows the minimum, maximum, median registration numbers of these five months, separately for online, offline and total number of registrations. The following additional facts are known.

1. In every month, both online and offline registration numbers were multiples of 10.
2. In January, the number of offline registrations was twice that of online registrations.
3. In April, the number of online registrations was twice that of offline registrations.
4. The number of online registrations in March was the same as the number of offline registrations in February.
5. The number of online registrations was the largest in May.

	Minimum	Maximum	Median
Online	100	100	80
Offline	30	80	50
Total	110	130	120

Handwritten notes on a whiteboard show calculations for Online, Offline, and Total registrations for each month (Jan to May) based on the given conditions.

VIDEO SOLUTION

Top-500 Free CAT Questions (With Solutions)

Instructions

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers - Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively. The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	1 and 4
Range of rating*	3	3	4	4	3

* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

- (a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.
- (b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

Question 25

How many individual ratings cannot be determined from the above information?

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers - Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively. The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	1 and 4
Range of rating*	3	3	4	4	3

* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

- (a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.
- (b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

Handwritten notes on a whiteboard show calculations for Mean and Median ratings for each worker based on the given information.

Instructions

Odsville has five firms - Alfloo, Bzygoo, Czechy, Drjbna and Elavalaki. Each of these firms was founded in some year and also closed down a few years later.

Each firm raised Rs. 1 crore in its first and last year of existence. The amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down. No firm raised the same amount of money in two consecutive years. Each annual increase and decrease was either by Rs. 1 crore or by Rs. 2 crores. The table below provides partial information about the five firms.

Firm	First year of existence	Last year of existence	Total amount raised (Rs. Crores)
Alfloo	2009	2016	21
Bzygoo	2012	2015	
Czechy	2013		9
Drjbna	2011	2015	10
Elavalaki	2010		13

Question 26

For which firm(s) can the amounts raised by them be concluded with certainty in each year?

- A Only Bzygoo and Czechy and Drjbna
- B Only Czechy and Drjbna
- C Only Drjbna
- D Only Czechy

Instructions

In an election several candidates contested for a constituency. In any constituency, the winning candidate was the one who polled the highest number of votes, the first runner up was the one who polled the second highest number of votes, the second runner up was the one who polled the third highest number of votes, and so on. There were no ties (in terms of number of votes polled by the candidates) in any of the constituencies in this election. In an electoral system, a security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election .

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	
No. of votes polled by the winning candidate	2,75,000	48,750		
No. of votes polled by the first runner up	95,000			37,500
No. of votes polled by the second runner up				30,000
% of valid votes polled by the third runner up				10%

The following additional facts are known:

1. The first runner up polled 10,000 more votes than the second runner up in constituency A.
2. None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
3. The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

Question 27

How many candidates who contested in constituency B lost their security deposit?

C Security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election.

	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	
No. of votes polled by the winning candidate	2,75,000	48,750		
No. of votes polled by the first runner up	95,000			37,500
No. of votes polled by the second runner up				30,000
% of valid votes polled by the third runner up				10%

The following additional facts are known:

1. The first runner up polled 10,000 more votes than the second runner up in constituency A.
2. None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
3. The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

Handwritten notes: $\text{Forfeits} = V \cdot \frac{1}{6} \cdot TV$. Circled numbers: 600, 90, 100, 101. Brackets and arrows indicating relationships between these numbers.

VIDEO SOLUTION

Free CAT Study Material

Instructions

The table given below shows the amount, in grams, of carbohydrate, protein, fat and all other nutrients, per 100 grams of nutrients in seven food grains. The first column shows the food grain category and the second column its codename. The table has some missing values.

Food grain Category	Codename of the food grain	Composition per hundred grams of nutrients in the food grains			
		Carbohydrate	Protein	Fat	Other nutrients
Cereal	C1			0	12
	C2			3	10
Millet	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereal	P1	66			10
	P2		14		8

The following additional facts are known.

1. Both the pseudo-cereals had higher amounts of carbohydrate as well as higher amounts of protein than any millet.

- Both the cereals had higher amounts of carbohydrate than any pseudo-cereal.
- All the missing values of carbohydrate amounts (in grams) for all the food grains are non-zero multiples of 5.
- All the missing values of protein, fat and other nutrients amounts (in grams) for all the food grains are non-zero multiples of 4.
- P1 contained double the amount of protein that M3 contains.

Question 28

What is the median of the number of grams of protein in 100 grams of nutrients among these food grains?

The table given below shows the amount, in grams, of carbohydrate, protein, fat and all other nutrients, per 100 grams of nutrients in seven food grains. The first column shows the food grain category and the second column its codename. The table has some missing values.

Food grain category	Codename	Carbohydrate	Protein	Fat	Other nutrients
Cereal	P1	66	14	2	1
Cereal	P2	66	14	2	1
Cereal	P3	66	14	2	1
Cereal	P4	66	14	2	1
Cereal	P5	66	14	2	1
Cereal	P6	66	14	2	1
Cereal	P7	66	14	2	1
Pseudo-cereal	M1	66	14	2	1
Pseudo-cereal	M2	66	14	2	1
Pseudo-cereal	M3	66	14	2	1
Pseudo-cereal	M4	66	14	2	1
Pseudo-cereal	M5	66	14	2	1
Pseudo-cereal	M6	66	14	2	1
Pseudo-cereal	M7	66	14	2	1

The following additional facts are known.

- Both the pseudo-cereals had higher amounts of carbohydrate than any higher amounts of protein than any millet.
- Both the cereals had higher amounts of carbohydrate than any pseudo-cereal.
- All the missing values of carbohydrate amounts (in grams) for all the food grains are non-zero multiples of 5.
- All the missing values of protein, fat and other nutrients amounts (in grams) for all the food grains are non-zero multiples of 4.
- P1 contained double the amount of protein that M3 contains.

VIDEO SOLUTION

Instructions

The year is 2089. Beijing, London, New York, and Paris are in contention to host the 2096 Olympics. The eventual winner is determined through several rounds of voting by members of the IOC with each member representing a different city. All the four cities in contention are also represented in IOC.

Round	Total Votes Cast	Maximum Votes Cast		Eliminated	
		City	No. of Votes	City	No. of Votes
1		London	30	New York	12
2	83	Paris	32	Beijing	21
3	75				

- In any round of voting, the city receiving the lowest number of votes in that round gets eliminated. The survivor after the last round of voting gets to host the event.
- A member is allowed to cast votes for at most two different cities in all rounds of voting combined. (Hence, a member becomes ineligible to cast a vote in a given round if both the city (s)he voted for in earlier rounds are out of contention in that round of voting.)
- A member is also ineligible to cast a vote in a round if the city (s)he represents is in contention in that round of voting.
- As long as the member is eligible, (s)he must vote and vote for only one candidate city in any round of voting. The following incomplete table shows the information on cities that received the maximum and minimum votes in different rounds, the number of votes cast in their favour, and the total votes that were cast in those rounds. It is also known that:
 - All those who voted for London and Paris in round 1, continued to vote for the same cities in subsequent rounds as long as these cities were in contention. 75% of those who voted for Beijing in round 1, voted for Beijing in round 2 as well.
 - Those who voted for New York in round 1, voted either for Beijing or Paris in round 2.
 - The difference in votes cast for the two contending cities in the last round was 1.
 - 50% of those who voted for Beijing in round 1, voted for Paris in round 3.

Question 29

What is the number of votes cast for Paris in round 1?

- A 16
- B 18
- C 22
- D 24



Read the following information carefully and answer the questions which follow.

The year is 2089. Beijing, London, New York, and Paris are in contention to host the 2096 Olympics. The eventual winner is determined through several rounds of voting by members of the IOC with each member representing a different city. All the four cities in contention are also represented in IOC.

In any round of voting, the city receiving the lowest number of votes in that round gets eliminated. The survivor after the last round of voting gets to host the event.

A member is allowed to vote for at most two different cities in all rounds of voting combined. (Hence, a member is ineligible to cast a vote in a given round if both the cities (s)he voted for are still in contention in that round of voting.) A member is also ineligible to cast a vote if the city (s)he represents is in contention in that round of voting. As long as a member is eligible, (s)he must vote and vote for only one candidate city in any round of voting. The following incomplete table shows the information on cities that received the maximum and minimum votes in different rounds, the number of votes cast in their favor, and the total votes that were cast in those rounds.

Round	Total votes	Maximum Votes Cast		Eliminated	
		City	No. of Votes	City	No. Of Votes
1		London	30	New York	12
2	83	Paris	32	Beijing	21
3	75				

 VIDEO SOLUTION

Instructions

Funky Pizzeria was required to supply pizzas to three different parties. The total number of pizzas it had to deliver was 800, 70% of which were to be delivered to Party 3 and the rest equally divided between Party 1 and Party 2.

Pizzas could be of Thin Crust (T) or Deep Dish (D) variety and come in either Normal Cheese (NC) or Extra Cheese (EC) versions. Hence, there are four types of pizzas: T-NC, T-EC, D-NC and D-EC. Partial information about proportions of T and NC pizzas ordered by the three parties is given below:

	Thin crust (T)	Normal cheese (NC)
Party 1	0.6	
Party 2	0.55	0.3
Party 3		0.65
Total	0.375	0.52

Question 30

How many Normal Cheese pizzas were required to be delivered to Party 1?

- A 104
- B 84
- C 16
- D 196

between Party 1 and Party 2

Pizzas could be of Thin Crust (T) or Deep Dish (D) variety and come in either Normal Cheese (NC) or Extra Cheese (EC) versions. Hence, there are four types of pizzas: T-NC, T-EC, D-NC and D-EC. Partial information about proportions of T and NC pizzas ordered by the three parties is given below:

	Thin crust (T)	Normal cheese (NC)
Party 1	0.6	0.4
Party 2	0.55	0.45
Party 3	0.375	0.625
Total		

Handwritten notes on the screen:

- $15/1 = 120$
- $15/1 = 120$
- $70/1 = 560$
- T-EC, D-EC, T-D, T-NC, D-NC, NC-EC
- Party 1
- Party 2
- Party 3

VIDEO SOLUTION

Free CAT Preparation Video Lectures

Question 31

How many Thin Crust pizzas were to be delivered to Party 3?

- A 398
- B 162
- C 96
- D 364

between Party 1 and Party 2

Pizzas could be of Thin Crust (T) or Deep Dish (D) variety and come in either Normal Cheese (NC) or Extra Cheese (EC) versions. Hence, there are four types of pizzas: T-NC, T-EC, D-NC and D-EC. Partial information about proportions of T and NC pizzas ordered by the three parties is given below:

	Thin crust (T)	Normal cheese (NC)
Party 1	0.6	0.4
Party 2	0.55	0.45
Party 3	0.375	0.625
Total		

Handwritten notes on the screen:

- $15/1 = 120$
- $15/1 = 120$
- $70/1 = 560$
- T-EC, D-EC, T-D, T-NC, D-NC, NC-EC
- Party 1
- Party 2
- Party 3

VIDEO SOLUTION

Instructions

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers -

Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	1 and 4
Range of rating*	3	3	4	4	3

* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

- (a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.
 (b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

Question 32

To how many workers did R2 give a rating of 4?

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers - Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	1 and 4
Range of rating*	3	3	4	4	3

* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

(a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.

(b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

VIDEO SOLUTION

Question 33

Which among the following restaurants gave its median rating to exactly one of the workers?

- A R2
- B R5
- C R4
- D R3

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers - Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	1 and 4
Range of rating*	3	3	4	4	3

* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

(a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.

(b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

Daily Free CAT Practice Tests

Instructions

Answer the following questions based on the information given below:

For admission to various affiliated colleges, a university conducts a written test with four different sections, each with a maximum of 50 marks. The following table gives the aggregate as well as the sectional cut-off marks fixed by six different colleges affiliated to the university. A student will get admission only if he/she gets marks greater than or equal to the cut-off marks in each of the sections and his/her aggregate marks are at least equal to the aggregate cut-off marks as specified by the college.

	Sectional Cut-Off Marks				Aggregate Cut - off Marks
	Section A	Section B	Section C	Section D	
College 1	42	42	42		176
College 2		45	45		175
College 3			46		171
College 4	43			45	178
College 5	45		43		180
College 6		41		44	176

Question 34

Aditya did not get a call from even a single college. What could be the maximum aggregate marks obtained by him?

- A 181
- B 176
- C 184
- D 196
- E 190

Answer the following questions based on the information given below:

For admission to various affiliated colleges, a university conducts a written test with four different sections, each with a maximum of 50 marks. The following table gives the aggregate as well as the sectional cut-off marks fixed by six different colleges affiliated to the university. A student will get admission only if he/she gets marks greater than or equal to the cut-off marks in each of the sections and his/her aggregate marks are at least equal to the aggregate cut-off marks as specified by the college.

	Section A	Section B	Section C	Section D	Aggregate Cut-off Marks
College 1	42	42	42		176
College 2		45	45		175
College 3			46		171
College 4	43			45	178
College 5	45		43		180
College 6		41		44	176

Bhama got calls from all colleges. What could be the minimum aggregate marks obtained by her?

A) 180
B) 181
C) 196
D) 176
E) 184

Handwritten notes: Section A: 5, Section B: 45, Section C: 46, Section D: 46. Total: 181.

Question 35

Charlie got calls from two colleges. What could be the minimum marks obtained by him in a section?

- A 0
- B 21

C 25

D 35

E 41

Answer the following questions based on the information given below:

For admission to various affiliated colleges, a university conducts a written test with four different sections, each with a maximum of 50 marks. The following table gives the aggregate as well as the sectional cut-off marks fixed by six different colleges affiliated to the university. A student will get admission only if he/she gets marks greater than or equal to the cut-off marks in each of the sections and his/her aggregate marks are at least equal to the aggregate cut-off marks as specified by the college.

College	Sectional Cut-Off Marks				Aggregate Cut-Off Marks
	Section A	Section B	Section C	Section D	
College 1	42	42	42	42	170
College 2	41	41	41	41	170
College 3	43	43	43	43	170
College 4	43	43	43	43	170
College 5	45	45	45	45	180
College 6	45	45	45	45	176

Bhama got calls from all colleges. What could be the minimum aggregate marks obtained by her?

A) 180
B) 181
C) 196
D) 176
E) 184

Handwritten notes: Section A: 5, Section B: 45, Section C: 46, Section D: 46. Bhama.



VIDEO SOLUTION

Question 36

Bhama got calls from all colleges. What could be the minimum aggregate marks obtained by her?

A 180

B 181

C 196

D 176

E 184

Answer the following questions based on the information given below:

For admission to various affiliated colleges, a university conducts a written test with four different sections, each with a maximum of 50 marks. The following table gives the aggregate as well as the sectional cut-off marks fixed by six different colleges affiliated to the university. A student will get admission only if he/she gets marks greater than or equal to the cut-off marks in each of the sections and his/her aggregate marks are at least equal to the aggregate cut-off marks as specified by the college.

College	Sectional Cut-Off Marks				Aggregate Cut-Off Marks
	Section A	Section B	Section C	Section D	
College 1	42	42	42	42	170
College 2	41	41	41	41	170
College 3	43	43	43	43	170
College 4	43	43	43	43	170
College 5	45	45	45	45	180
College 6	45	45	45	45	176

Bhama got calls from all colleges. What could be the minimum aggregate marks obtained by her?

A) 180
B) 181
C) 196
D) 176
E) 184

Handwritten notes: Section A: 5, Section B: 45, Section C: 46, Section D: 46. Bhama.



VIDEO SOLUTION

Free Videos for CAT Preparation

Instructions

There are only three female students - Amala, Koli and Rini - and only three male students - Biman, Mathew and Shyamal - in a course. The course has two evaluation components, a project and a test. The aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

The projects are done in groups of two, with each group consisting of a female and a male student. Both the group members obtain the same score in the project.

The following additional facts are known about the scores in the project and the test.

1. The minimum, maximum and the average of both project and test scores were identical - 40, 80 and 60, respectively.
2. The test scores of the students were all multiples of 10; four of them were distinct and the remaining two were equal to the average test scores.
3. Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Yet Amala had the highest aggregate score.
4. Shyamal scored the second highest in the test. He scored two more than Koli, but two less than Amala in the aggregate.
5. Biman scored the second lowest in the test and the lowest in the aggregate.
6. Mathew scored more than Rini in the project, but less than her in the test.

Question 37

What was Rini's score in the project?

There are only three female students - Amala, Koli and Rini - and only three male students - Biman, Mathew and Shyamal - in a course. The course has two evaluation components, a project and a test. The aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

The projects are done in groups of two, with each group consisting of a female and a male student. Both the group members obtain the same score in the project.

The following additional facts are known about the scores in the project and the test.

1. The minimum, maximum and the average of both project and test scores were identical - 40, 80 and 60, respectively.
2. The test scores of the students were all multiples of 10; four of them were distinct and the remaining two were equal to the average test scores.
3. Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Yet Amala had the highest aggregate score.
4. Shyamal scored the second highest in the test. He scored two more than Koli, but two less than Amala in the aggregate.
5. Biman scored the second lowest in the test and the lowest in the aggregate.
6. Mathew scored more than Rini in the project, but less than her in the test.

VIDEO SOLUTION

Instructions

A company administers a written test comprising of three sections of 20 marks each - Data Interpretation (DI), Written English (WE) and General Awareness (GA), for recruitment. A composite score for a candidate (out of 80) is calculated by doubling her marks in DI and adding it to the sum of her marks in the other two sections. Candidates who score less than 70% marks in two or more sections are disqualified. From among the rest, the four with the highest composite scores are recruited. If four or less candidates qualify, all who qualify are recruited.

Ten candidates appeared for the written test. Their marks in the test are given in the table below. Some marks in the table are missing, but the following facts are known:

1. No two candidates had the same composite score.
2. Ajay was the unique highest scorer in WE.
3. Among the four recruited, Geeta had the lowest composite score.
4. Indu was recruited.
5. Danish, Harini, and Indu had scored the same marks in GA.
6. Indu and Jatin both scored 100% in exactly one section and Jatin's composite score was 10 more than Indu's.

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	
No. of votes polled by the winning candidate	2,75,000	48,750		
No. of votes polled by the first runner up	95,000			37,500
No. of votes polled by the second runner up				30,000
% of valid votes polled by the third runner up				10%

The following additional facts are known:

1. The first runner up polled 10,000 more votes than the second runner up in constituency A.
2. None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
3. The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

Question 39

What is the percentage of votes polled in total by all the candidates who lost their security deposits while contesting for constituency A?

Security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election.

	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	
No. of votes polled by the winning candidate	2,75,000	48,750		
No. of votes polled by the first runner up	95,000			37,500
No. of votes polled by the second runner up				30,000
% of valid votes polled by the third runner up				10%

The following additional facts are known:

1. The first runner up polled 10,000 more votes than the second runner up in constituency A.
2. None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
3. The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

Handwritten notes: Forfeit = $V \cdot \leq \frac{1}{6} \cdot TV$.
 $600 \cdot 90 \cdot 100 \leq 101$



VIDEO SOLUTION

CAT Percentile Predictor

Instructions

A company administers a written test comprising of three sections of 20 marks each - Data Interpretation (DI), Written English (WE) and General Awareness (GA), for recruitment. A composite score for a candidate (out of 80) is calculated by doubling her marks in DI and adding it to the sum of her marks in the other two sections. Candidates who score less than 70% marks in two or more sections are disqualified. From among the rest, the four with the highest composite scores are recruited. If four or less candidates qualify, all who qualify are recruited.

Ten candidates appeared for the written test. Their marks in the test are given in the table below. Some marks in the table are missing, but the following facts are known:

1. No two candidates had the same composite score.
2. Ajay was the unique highest scorer in WE.
3. Among the four recruited, Geeta had the lowest composite score.
4. Indu was recruited.
5. Danish, Harini, and Indu had scored the same marks in GA.
6. Indu and Jatin both scored 100% in exactly one section and Jatin's composite score was 10 more than Indu's.

Candidate	Marks out of 20		
	DI	WE	GA
Ajay	8		16
Bala		9	11
Chetna	19	4	12
Danish	8	15	
Ester	12	18	16
Falak	15	7	10
Geeta	14		6
Harini	5		
Indu		8	
Jatin		16	14

Question 40

If all the candidates except Ajay and Danish had different marks in DI, and Bala's composite score was less than Chetna's composite score, then what is the maximum marks that Bala could have scored in DI?

- No two candidates had the same composite score.
- Ajay was the unique highest scorer in WE.
- Among the four recruited, Geeta had the lowest composite score.
- Indu was recruited.
- Danish, Harini, and Indu had scored the same marks in WE.
- Indu and Jatin both scored 100% in exactly one section and Jatin's composite score was 10 more than Indu's.

Candidate	DI	WE	GA
Ajay	8		16
Bala		9	11
Chetna	19	4	12
Danish	8	15	
Ester	12	18	16
Falak	15	7	10
Geeta	14		6
Harini	5		
Indu		8	
Jatin		16	14

VIDEO SOLUTION

Instructions

Odsville has five firms - Alfloo, Bzygoo, Czechy, Drjbna and Elavalaki. Each of these firms was founded in some year and also closed down a few years later.

Each firm raised Rs. 1 crore in its first and last year of existence. The amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down. No firm raised the same amount of money in two consecutive years. Each annual increase and decrease was either by Rs. 1 crore or by Rs. 2 crores. The table below provides partial information about the five firms.

Firm	First year of existence	Last year of existence	Total amount raised (Rs. Crores)
Alfloo	2009	2016	21
Bzygoo	2012	2015	
Czechy	2013		9
Drjbna	2011	2015	10
Elavalaki	2010		13

Question 41

What best can be concluded about the total amount of money raised in 2015?

- A It is either Rs. 7 crores or Rs. 8 crores or Rs. 9 crores.
- B It is exactly Rs. 8 crores.

C It is either Rs. 7 crores or Rs. 8 crores.

D It is either Rs. 8 crores or Rs. 9 crores.

cracku

Odville has five firms - Ailhos, Bzygyn, Crcchy, Orljns and Elavalaki. Each of these firms was founded in some year and also closed down a few years later.

Each firm raised Rs. 1 crore in its first and last year of existence. The amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down. No firm raised the same amount of money in two consecutive years. Each annual increase and decrease was either by Rs. 1 crore or by Rs. 2 crores.

The table below provides partial information about the five firms.

Firm	First year of existence	Last year of existence	Total amount raised (Rs. Crores)
Ailhos	2009	2010	21
Bzygyn	2012	2015	0
Crcchy	2013	2014	10
Orljns	2014	2016	11

VIDEO SOLUTION

Question 42

If Elavalaki raised Rs. 3 crores in 2013, then what is the smallest possible total amount of money (in Rs. crores) that could have been raised by all the companies in 2012?

A 12

B 9

C 11

D 10

cracku

Odville has five firms - Ailhos, Bzygyn, Crcchy, Orljns and Elavalaki. Each of these firms was founded in some year and also closed down a few years later.

Each firm raised Rs. 1 crore in its first and last year of existence. The amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down. No firm raised the same amount of money in two consecutive years. Each annual increase and decrease was either by Rs. 1 crore or by Rs. 2 crores.

The table below provides partial information about the five firms.

Firm	First year of existence	Last year of existence	Total amount raised (Rs. Crores)
Ailhos	2009	2010	21
Bzygyn	2012	2015	0
Crcchy	2013	2014	10
Orljns	2014	2016	11

VIDEO SOLUTION

Important Verbal Ability Questions for CAT (Download PDF)

Instructions

In a coaching class, some students register online, and some others register offline. No student registers both online and offline; hence the total registration number is the sum of online and offline registrations. The following facts and table pertain to these registration numbers for the five months - January to May of 2023. The table shows the minimum, maximum, median registration numbers of these five months, separately for online, offline and total number of registrations. The following additional facts are known.

1. In every month, both online and offline registration numbers were multiples of 10.
2. In January, the number of offline registrations was twice that of online registrations.
3. In April, the number of online registrations was twice that of offline registrations.
4. The number of online registrations in March was the same as the number of offline registrations in February.
5. The number of online registrations was the largest in May.

	Minimum	Maximum	Median
Online	40	100	80
Offline	30	80	50
Total	110	130	120

Question 43

Which of the following statements can be true?

- I. The number of offline registrations was the smallest in May.
- II. The total number of registrations was the smallest in February.

- A Both I and II
- B Only II
- C Neither I nor II
- D Only I

shows the minimum, maximum, median registration numbers of these five months, separately for online, offline and total number of registrations. The following additional facts are known.

1. In every month, both online and offline registration numbers were multiples of 10.
2. In January, the number of offline registrations was twice that of online registrations.
3. In April, the number of online registrations was twice that of offline registrations.
4. The number of online registrations in March was the same as the number of offline registrations in February.
5. The number of online registrations was the largest in May.

	Minimum	Maximum	Median
Online	40	100	80
Offline	30	80	50
Total	110	130	120

VIDEO SOLUTION

Instructions

The table given below shows the amount, in grams, of carbohydrate, protein, fat and all other nutrients, per 100 grams of nutrients in seven food grains. The first column shows the food grain category and the second column its codename. The table has some missing values.

Food grain Category	Codename of the food grain	Composition per hundred grams of nutrients in the food grains			
		Carbohydrate	Protein	Fat	Other nutrients
Cereal	C1			0	12
	C2			3	10
Millet	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereal	P1	66			10
	P2		14		8

The following additional facts are known.

1. Both the pseudo-cereals had higher amounts of carbohydrate as well as higher amounts of protein than any millet.
2. Both the cereals had higher amounts of carbohydrate than any pseudo-cereal.
3. All the missing values of carbohydrate amounts (in grams) for all the food grains are non-zero multiples of 5.
4. All the missing values of protein, fat and other nutrients amounts (in grams) for all the food grains are non-zero multiples of 4.
5. P1 contained double the amount of protein that M3 contains.

Question 44

How many foodgrains had a higher amount of carbohydrate per 100 grams of nutrients than M1?

VIDEO SOLUTION

Instructions

In an election several candidates contested for a constituency. In any constituency, the winning candidate was the one who polled the highest number of votes, the first runner up was the one who polled the second highest number of votes, the second runner up was the one who polled the third highest number of votes, and so on. There were no ties (in terms of number of votes polled by the candidates) in any of the constituencies in this election. In an electoral system, a security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election .

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	
No. of votes polled by the winning candidate	2,75,000	48,750		
No. of votes polled by the first runner up	95,000			37,500
No. of votes polled by the second runner up				30,000
% of valid votes polled by the third runner up				10%

The following additional facts are known:

1. The first runner up polled 10,000 more votes than the second runner up in constituency A.
2. None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
3. The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

Question 45

What BEST can be concluded about the number of votes polled by the winning candidate in constituency C?

- A 1,40,010
- B between 1,40,005 and 1,40,010
- C less than 2,00,010
- D 1,40,006

Security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election.

	Constituency			
	A	B	C	D
No. of candidates contesting	10	15	5	—
Total no. of valid votes polled	5,00,000	3,25,000	6,00,000	—
No. of votes polled by the winning candidate	2,75,000	48,750	—	37,500
No. of votes polled by the first runner up	95,000	—	—	30,000
No. of votes polled by the second runner up	—	—	—	10,000
% of valid votes polled by the third runner up	—	—	—	10%

The following additional facts are known:

- The first runner up polled 10,000 more votes than the second runner up in constituency A.
- None of the candidates who contested in constituency lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
- The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

VIDEO SOLUTION

[Data Interpretation for CAT Questions \(download pdf\)](#)

Question 46

The winning margin of a constituency is defined as the difference of votes polled by the winner and that of the first runner up. Which of the following CANNOT be the list of constituencies, in increasing order of winning margin?

- A D, B, C, A
- B B, D, C, A
- C B, C, D, A
- D D, C, B, A

Security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election.

	Constituency			
	A	B	C	D
No. of candidates contesting	10	15	5	—
Total no. of valid votes polled	5,00,000	3,25,000	6,00,000	—
No. of votes polled by the winning candidate	2,75,000	48,750	—	37,500
No. of votes polled by the first runner up	95,000	—	—	30,000
No. of votes polled by the second runner up	—	—	—	10,000
% of valid votes polled by the third runner up	—	—	—	10%

The following additional facts are known:

- The first runner up polled 10,000 more votes than the second runner up in constituency A.
- None of the candidates who contested in constituency lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
- The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

VIDEO SOLUTION

Instructions

An agency entrusted to accredit colleges looks at four parameters: faculty quality (F), reputation (R), placement quality (P), and infrastructure (I). The four parameters are used to arrive at an overall score, which the agency uses to give an accreditation to the colleges. In each parameter, there are five possible letter grades given, each carrying certain points: A (50 points), B (40 points), C (30 points), D (20 points), and F (0 points). The overall score for a college is the weighted sum of the points scored in the four parameters. The weights of the parameters are 0.1, 0.2, 0.3 and 0.4 in some order, but the order is not disclosed. Accreditation is awarded based on the following scheme:

Range	Accreditation
Overall score ≥ 45	AAA
$35 \leq \text{Overall score} < 45$	BAA
$25 \leq \text{Overall score} < 35$	BBA
$15 \leq \text{Overall score} < 25$	BBB
Overall score < 15	Junk

Eight colleges apply for accreditation, and receive the following grades in the four parameters (F, R, P, and I):

	F	R	P	I
A-one	A	A	A	B
Best Ed	B	C	D	D
Cosmopolitan	B	D	D	C
Dominance	D	D	B	C
Education Aid	A	A	B	A
Fancy	A	A	B	B
Global	C	F	D	D
High Q	C	D	D	B

It is further known that in terms of overall scores:

1. High Q is better than Best Ed;
2. Best Ed is better than Cosmopolitan; and
3. Education Aid is better than A-one.

Question 47

What is the weight of the faculty quality parameter?

- A 0.2
- B 0.3
- C 0.4
- D 0.1

Eight colleges apply for accreditation, and receive the following grades in the four parameters (F, R, P, and I):

	F	R	P	I
A-one	A	A	A	B
Best Ed	B	C	D	D
Cosmopolitan	B	D	D	C
Dominance	D	D	B	C
Education Aid	A	A	B	A
Fancy	A	A	B	B
Global	C	F	D	D
High Q	C	D	D	B

It is further known that in terms of overall scores:

1. High Q is better than Best Ed;
2. Best Ed is better than Cosmopolitan; and
3. Education Aid is better than A-one.

Handwritten notes in the video show the calculation of overall scores for each college:

- A-one: $50 \times 0.1 + 50 \times 0.2 + 50 \times 0.3 + 40 \times 0.4 = 45$
- Best Ed: $40 \times 0.1 + 30 \times 0.2 + 20 \times 0.3 + 20 \times 0.4 = 35$
- Cosmopolitan: $40 \times 0.1 + 20 \times 0.2 + 20 \times 0.3 + 30 \times 0.4 = 35$
- Dominance: $20 \times 0.1 + 20 \times 0.2 + 40 \times 0.3 + 30 \times 0.4 = 35$
- Education Aid: $50 \times 0.1 + 50 \times 0.2 + 40 \times 0.3 + 50 \times 0.4 = 40$
- Fancy: $50 \times 0.1 + 50 \times 0.2 + 40 \times 0.3 + 40 \times 0.4 = 40$
- Global: $30 \times 0.1 + 0 \times 0.2 + 20 \times 0.3 + 20 \times 0.4 = 20$
- High Q: $30 \times 0.1 + 20 \times 0.2 + 20 \times 0.3 + 40 \times 0.4 = 35$

VIDEO SOLUTION

Question 48

How many colleges have overall scores between 31 and 40, both inclusive?

- A 0
- B 2
- C 1
- D 3

VIDEO SOLUTION

Logical Reasoning for CAT Questions (download pdf)

Instructions

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of these people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran		1	
Dhinesh			2
Ezhil			1
Fani	1		
Gowtham		3	2

The following additional facts are also known.

- Amudha, Bharatan, and Chandran are mutual strangers.
- Amudha, Dhinesh, and Fani are Ezhil's friends.
- Chandran and Gowtham are friends.
- Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
- Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

Question 49

Which of these pairs share the same type of relationship?

- A** (Amudha, Gowtham) and (Ezhil, Fani)
- B** (Bharatan, Chandran) and (Dhinesh, Ezhil)
- C** (Chandran, Ezhil) and (Dhinesh, Gowtham)
- D** (Bharatan, Ezhil) and (Fani, Gowtham)

cracku

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of the seven people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran	1		2
Dhinesh			1
Ezhil	3		2
Fani			
Gowtham			

Following additional facts are also known:

- Amudha, Bharatan, and Chandran are mutual strangers.
- Amudha, Dhinesh, and Fani are Ezhil's friends.
- Chandran and Gowtham are friends.
- Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
- Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

VIDEO SOLUTION

Question 50

Who is an acquaintance of Chandran?

- A** Dhinesh
- B** Fani
- C** Ezhil
- D** Bharatan

cracku

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of the seven people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran	1		2
Dhinesh			1
Ezhil	3		2
Fani			
Gowtham			

Following additional facts are also known:

- Amudha, Bharatan, and Chandran are mutual strangers.
- Amudha, Dhinesh, and Fani are Ezhil's friends.
- Chandran and Gowtham are friends.
- Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
- Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

VIDEO SOLUTION

Instructions

Odsville has five firms - Alfloo, Bzygoo, Czechy, Drijbna and Elavalaki. Each of these firms was founded in some year and also closed down a few years later.

Each firm raised Rs. 1 crore in its first and last year of existence. The amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down. No firm raised the same amount of money in two consecutive years. Each annual increase and decrease was either by Rs. 1 crore or by

Rs. 2 crores. The table below provides partial information about the five firms.

Firm	First year of existence	Last year of existence	Total amount raised (Rs. Crores)
Alfloo	2009	2016	21
Bzygoo	2012	2015	
Czechy	2013		9
Drjbna	2011	2015	10
Elavalaki	2010		13

Question 51

If the total amount of money raised in 2014 is Rs. 12 crores, then which of the following is not possible?

- A Bzygoo raised the same amount of money as Elavalaki in 2013.
- B Alfloo raised the same amount of money as Drjbna in 2013.
- C Alfloo raised the same amount of money as Bzygoo in 2014.
- D Bzygoo raised more money than Elavalaki in 2014.

VIDEO SOLUTION

Quantitative Aptitude for CAT Questions (download pdf)

Instructions

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of these people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran		1	
Dhinesh			2
Ezhil			1
Fani	1		
Gowtham		3	2

The following additional facts are also known.

1. Amudha, Bharatan, and Chandran are mutual strangers.
2. Amudha, Dhinesh, and Fani are Ezil's friends.
3. Chandran and Gowtham are friends.
4. Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
5. Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

Question 52

Who are Gowtham's acquaintances?

- A Dhinesh, Ezhil and Fani
- B Amudha, Dhinesh and Fani
- C Bharatan, Dhinesh and Ezhil
- D Amudha, Bharatan and Fani

cracku

	A	B	C	D	E	F	G
A							
B							
C							
D							
E							
F							
G							

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of the people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha	1		4
Bharatan		1	
Chandran			2
Dhinesh	3		2

The following additional facts are also known.

1. Amudha, Bharatan, and Chandran are mutual strangers.
2. Amudha, Dhinesh, and Fani are Ezil's friends.
3. Chandran and Gowtham are friends.
4. Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
5. Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

VIDEO SOLUTION

Instructions

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers -

Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	1 and 4
Range of rating*	3	3	4	4	3

* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

- (a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.
- (b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

Question 53

What rating did R1 give to Xavier?

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers - Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	1 and 4
Range of rating*	3	3	4	4	3

* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

(a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.

(b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

Handwritten notes on a whiteboard show the calculation of the mean rating for R1: $1, 2, 3, 4, 5$ and $\text{Mean} = \frac{1+2+3+4+5}{5} = 3$. The median is also noted as 3. A video player interface is overlaid on the image.

VIDEO SOLUTION

Instructions

DIRECTIONS for the following four questions:

The proportion of male students and the proportion of vegetarian students in a school are given below. The school has a total of 800 students, 80% of whom are in the Secondary Section and rest equally divided between Class 11 and 12.

	Male (M)	Vegetarian (V)
Class 12	0.6	
Class 11	0.55	0.5
Secondary Section		0.55
Total	0.475	0.53

Question 54

In the Secondary Section, 50% of the males are vegetarian. Which of the following statements is correct?

- A Except vegetarian males, all other groups have same number of students.
- B Except non-vegetarian males, all other groups have same number of students.
- C Except vegetarian females, all other groups have same number of students.
- D Except non-vegetarian females, all other groups have same number of students.
- E All of the above groups have the same number of students.

VIEW SOLUTION

Know the CAT Percentile Required for IIM Calls

Question 55

What is the percentage of vegetarian students in Class 12?

- A 40
- B 45
- C 50
- D 55

E 60

[VIEW SOLUTION](#)

Question 56

In Class 12, twenty five per cent of the vegetarians are male. What is the difference between the number of female vegetarians and male non-vegetarians?

- A less than 8
- B 10
- C 12
- D 14
- E 16

[VIEW SOLUTION](#)

Question 57

What is the percentage of male students in the secondary section?

- A 40
- B 45
- C 50
- D 55
- E 60

[VIEW SOLUTION](#)

[Join MBA Telegram Group](#)

Instructions

The table given below shows the amount, in grams, of carbohydrate, protein, fat and all other nutrients, per 100 grams of nutrients in seven food grains. The first column shows the food grain category and the second column its codename. The table has some missing values.

Food grain Category	Codename of the food grain	Composition per hundred grams of nutrients in the food grains			
		Carbohydrate	Protein	Fat	Other nutrients
Cereal	C1			0	12
	C2			3	10
Millet	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereal	P1	66			10
	P2		14		8

The following additional facts are known.

1. Both the pseudo-cereals had higher amounts of carbohydrate as well as higher amounts of protein than any millet.

- Both the cereals had higher amounts of carbohydrate than any pseudo-cereal.
- All the missing values of carbohydrate amounts (in grams) for all the food grains are non-zero multiples of 5.
- All the missing values of protein, fat and other nutrients amounts (in grams) for all the food grains are non-zero multiples of 4.
- P1 contained double the amount of protein that M3 contains.

Question 58

How many grams of protein were there in 100 grams of nutrients in M2?

The table given below shows the amount, in grams, of carbohydrate, protein, fat and all other nutrients, per 100 grams of nutrients in seven food grains. The first column shows the food grain category and the second column its codename. The table has some missing values.

Food grain category	Codename	Carbohydrate	Protein	Fat	Other nutrients
Cereal	C1	66	14	2	1
Cereal	C2	66	14	2	1
Pseudo-cereal	P1	66	14	2	1
Pseudo-cereal	P2	66	14	2	1
Food grain	M1	66	14	2	1
Food grain	M2	66	14	2	1
Food grain	M3	66	14	2	1

The following additional facts are known.

- Both the pseudo-cereals had higher amounts of carbohydrate than any cereal.
- Both the cereals had higher amounts of carbohydrate than any pseudo-cereal.
- All the missing values of carbohydrate amounts (in grams) for all the food grains are non-zero multiples of 5.
- All the missing values of protein, fat and other nutrients amounts (in grams) for all the food grains are non-zero multiples of 4.
- P1 contained double the amount of protein that M3 contains.

VIDEO SOLUTION

Instructions

The year is 2089. Beijing, London, New York, and Paris are in contention to host the 2096 Olympics. The eventual winner is determined through several rounds of voting by members of the IOC with each member representing a different city. All the four cities in contention are also represented in IOC.

Round	Total Votes Cast	Maximum Votes Cast		Eliminated	
		City	No. of Votes	City	No. of Votes
1		London	30	New York	12
2	83	Paris	32	Beijing	21
3	75				

- In any round of voting, the city receiving the lowest number of votes in that round gets eliminated. The survivor after the last round of voting gets to host the event.
- A member is allowed to cast votes for at most two different cities in all rounds of voting combined. (Hence, a member becomes ineligible to cast a vote in a given round if both the city (s)he voted for in earlier rounds are out of contention in that round of voting.)
- A member is also ineligible to cast a vote in a round if the city (s)he represents is in contention in that round of voting.
- As long as the member is eligible, (s)he must vote and vote for only one candidate city in any round of voting. The following incomplete table shows the information on cities that received the maximum and minimum votes in different rounds, the number of votes cast in their favour, and the total votes that were cast in those rounds. It is also known that:
 - All those who voted for London and Paris in round 1, continued to vote for the same cities in subsequent rounds as long as these cities were in contention. 75% of those who voted for Beijing in round 1, voted for Beijing in round 2 as well.
 - Those who voted for New York in round 1, voted either for Beijing or Paris in round 2.
 - The difference in votes cast for the two contending cities in the last round was 1.
 - 50% of those who voted for Beijing in round 1, voted for Paris in round 3.

Question 59

What percentage of members from among those who voted for Beijing in round 2 and were eligible to vote in round 3, voted for London?

- A 33.33
- B 38.10
- C 50
- D 66.67

cracku

Read the following information carefully and answer the questions which follow.

The year is 2089. Beijing, London, New York, and Paris are in contention to host the 2096 Olympics. The eventual winner is determined through several rounds of voting by members of the IOC with each member representing a different city. All the four cities in contention are also represented in IOC.

In any round of voting, the city receiving the lowest number of votes in that round gets eliminated. The survivor after the last round of voting gets to host the event.

A member is allowed to vote for at most two different cities in all rounds of voting combined. (Hence, a member is ineligible to cast a vote in a given round if both the cities (s)he voted for are out of contention in that round of voting.) A member is also ineligible to cast a vote in a round if the city (s)he represents is in contention in that round of voting. As long as a member is eligible, (s)he must vote and vote for only one candidate city in any round of voting. The following incomplete table shows the information on cities that received the maximum and minimum votes in different rounds, the number of votes cast in their favour, and the total votes that were cast in those rounds.

Round	Total votes	Maximum Votes Cast		Eliminated	
		City	No. of Votes	City	No. Of Votes
1		London	30	New York	12
2	83	Paris	32	Beijing	21
3	75				

VIDEO SOLUTION

Instructions

In an election several candidates contested for a constituency. In any constituency, the winning candidate was the one who polled the highest number of votes, the first runner up was the one who polled the second highest number of votes, the second runner up was the one who polled the third highest number of votes, and so on. There were no ties (in terms of number of votes polled by the candidates) in any of the constituencies in this election. In an electoral system, a security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election .

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	
No. of votes polled by the winning candidate	2,75,000	48,750		
No. of votes polled by the first runner up	95,000			37,500
No. of votes polled by the second runner up				30,000
% of valid votes polled by the third runner up				10%

The following additional facts are known:

- The first runner up polled 10,000 more votes than the second runner up in constituency A.
- None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
- The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

Question 60

For all the four constituencies taken together, what was the approximate number of votes polled by all the candidates who lost their security deposit expressed as a percentage of the total valid votes from these four constituencies?

- A 38.25%
- B 23.54%
- C 23.91%
- D 32.00%

C Security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election.

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	8	11
Total no. of valid votes polled	3,00,000	3,25,000	4,00,000	3,75,000
No. of votes polled by the winning candidate	2,75,000	48,750	37,500	30,000
No. of votes polled by the first runner up	95,000			
No. of votes polled by the second runner up				
% of valid votes polled by the third runner up			10%	

The following additional facts are known:

1. The first runner up polled 10,000 more votes than the second runner up in constituency A.
2. None of the candidates who contested in constituency D lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
3. The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

Handwritten notes: $\text{Forfeited} = V \cdot \leq \frac{1}{6} \cdot TV$. Diagrams showing vote distributions: (600), (90), {, } and (101).



VIDEO SOLUTION

Enroll for Excellent CAT/MBA courses

Question 61

What was the number of valid votes polled in constituency D?

- A 1,25,000
- B 1,50,000
- C 1,75,000
- D 62,500

C Security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election.

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	8	11
Total no. of valid votes polled	3,00,000	3,25,000	4,00,000	3,75,000
No. of votes polled by the winning candidate	2,75,000	48,750	37,500	30,000
No. of votes polled by the first runner up	95,000			
No. of votes polled by the second runner up				
% of valid votes polled by the third runner up			10%	

The following additional facts are known:

1. The first runner up polled 10,000 more votes than the second runner up in constituency A.
2. None of the candidates who contested in constituency D lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
3. The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

Handwritten notes: $\text{Forfeited} = V \cdot \leq \frac{1}{6} \cdot TV$. Diagrams showing vote distributions: (600), (90), {, } and (101).



VIDEO SOLUTION

Instructions

A chain of departmental stores has outlets in Delhi, Mumbai, Bengaluru and Kolkata. The sales are categorized by its three departments - 'Apparel', 'Electronics', and 'Home Décor'. An Accountant has been asked to prepare a summary of the 2018 and 2019 sales amounts for an internal report. He has collated partial information and prepared the following table.

	Sales Amounts (Crore Rupees)							
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparels	-	-	-	-	-	-	-	54
Electronics	78	98	82	102	90	70	80	100
HomeDecor	-	100	-	72	-	80	-	54

The following additional information is known.

1. The sales amounts in the Apparel departments were the same for Delhi and Kolkata in 2018.
2. The sales amounts in the Apparel departments were the same for Mumbai and Bengaluru in 2018. This sales amount matched the sales amount in the Apparel department for Delhi in 2019.
3. The sales amounts in the Home Décor departments were the same for Mumbai and Kolkata in 2018.
4. The sum of the sales amounts of four Electronics departments increased by the same amount as the sum of the sales amounts of four Apparel departments from 2018 to 2019.
5. The total sales amounts of the four Home Décor departments increased by Rs 70 Crores from 2018 to 2019.
6. The sales amounts in the Home Décor departments of Delhi and Bengaluru each increased by Rs 20 Crores from 2018 to 2019.
7. The sales amounts in the Apparel departments of Delhi and Bengaluru each increased by the same amount in 2019 from 2018. The sales amounts in the Apparel departments of Mumbai and Kolkata also each increased by the same amount in 2019 from 2018.
8. The sales amounts in the Apparel departments of Delhi, Kolkata and Bengaluru in 2019 followed an Arithmetic Progression.

Question 62

What was the total sales amount, in Crore Rupees, in 2019 for the chain of departmental stores?

- A 600
- B 900
- C 150
- D 750

VIDEO SOLUTION

Question 63

Among all the 12 departments (i.e., the 3 departments in each of the 4 cities), what was the maximum percentage increase in sales amount from 2018 to 2019?

- A 25
- B 28
- C 75
- D 50

Cracku CAT Success Stories

Accountant has been asked to prepare a summary of the 2018 and 2019 sales amounts for an internal report. He has collated partial information and prepared the following table.

	Mumbai		Delhi		Bengaluru	
	2018	2019	2018	2019	2018	2019
Apparel	25	28	25	28	25	28
Electronics	75	80	75	80	75	80
Home Décor	50	55	50	55	50	55

The following additional information is known.

1. The sales amounts in the Apparel departments were the same for Delhi and Kolkata in 2018.
2. The sales amounts in the Apparel departments were the same for Mumbai and Bengaluru in 2018. This sales amount was the sales amount in the Apparel department for Delhi in 2019.
3. The sales amounts in the Home Décor departments were the same for Mumbai and Kolkata in 2018.
4. The sum of the sales amounts of four Electronics departments increased by the same amount as the sum of the sales amount of four Apparel departments from 2018 to 2019.
5. The total sales amounts of the four Home Décor departments increased by Rs 70 Crores from 2018 to 2019.
6. The sales amounts in the Home Décor departments of Delhi and Bengaluru each increased by Rs 20 Crores from 2018 to 2019.
7. The sales amounts in the Apparel departments of Delhi and Bengaluru each increased by the same amount in 2019 from 2018.

VIDEO SOLUTION

Cracku CAT Success Stories

Question 64

What was the increase in sales amount, in Crore Rupees, in the Apparel department of Mumbai from 2018 to 2019?

- A 12
- B 8
- C 5
- D 10

Cracku CAT Success Stories

Accountant has been asked to prepare a summary of the 2018 and 2019 sales amounts for an internal report. He has collated partial information and prepared the following table.

	Mumbai		Delhi		Bengaluru	
	2018	2019	2018	2019	2018	2019
Apparel	25	28	25	28	25	28
Electronics	75	80	75	80	75	80
Home Décor	50	55	50	55	50	55

The following additional information is known.

1. The sales amounts in the Apparel departments were the same for Delhi and Kolkata in 2018.
2. The sales amounts in the Apparel departments were the same for Mumbai and Bengaluru in 2018. This sales amount was the sales amount in the Apparel department for Delhi in 2019.
3. The sales amounts in the Home Décor departments were the same for Mumbai and Kolkata in 2018.
4. The sum of the sales amounts of four Electronics departments increased by the same amount as the sum of the sales amount of four Apparel departments from 2018 to 2019.
5. The total sales amounts of the four Home Décor departments increased by Rs 70 Crores from 2018 to 2019.
6. The sales amounts in the Home Décor departments of Delhi and Bengaluru each increased by Rs 20 Crores from 2018 to 2019.
7. The sales amounts in the Apparel departments of Delhi and Bengaluru each increased by the same amount in 2019 from 2018.

VIDEO SOLUTION

Question 65

In Home Décor departments of which cities were the sales amounts the highest in 2018 and 2019, respectively?

- A Bengaluru and Delhi
- B Mumbai and Mumbai
- C Mumbai and Delhi
- D Delhi and Delhi

Cracku

Question 66

Bengaluru and Kolkata. The sales are categorized by its three departments - Apparel, Electronics, and Home Décor. An Accountant has been asked to prepare a summary of the 2018 and 2019 sales amounts for an internal report. He has collated partial information and prepared the following table.

	Bengaluru		Kolkata	
	2018	2019	2018	2019
Apparel	80	120	50	70
Electronics	100	150	100	150
Home Décor	120	180	120	180

The following additional information is known.

1. The sales amounts in the Apparel departments were the same for Delhi and Kolkata in 2018.
2. The sales amounts in the Apparel departments were the same for Mumbai and Bengaluru in 2018. This sales amount was the same as the sales amount in the Apparel department for Delhi in 2019.
3. The sales amounts in the Home Décor departments were the same for Mumbai and Kolkata in 2018.
4. The sum of the sales amounts of four Electronics departments increased by the same amount as the sum of the sales amounts of four Apparel departments from 2018 to 2019.
5. The total sales amounts of the four Home Décor departments increased by Rs 70 Crores from 2018 to 2019.
6. The sales amounts in the Home Décor departments of Delhi and Bengaluru each increased by Rs 20 Crores from 2018 to 2019.
7. The sales amounts in the Apparel departments of Delhi and Bengaluru each increased by the same amount in 2019 from 2018.

VIDEO SOLUTION

Instructions

The table given below shows the amount, in grams, of carbohydrate, protein, fat and all other nutrients, per 100 grams of nutrients in seven food grains. The first column shows the food grain category and the second column its codename. The table has some missing values.

Food grain Category	Codename of the food grain	Composition per hundred grams of nutrients in the food grains			
		Carbohydrate	Protein	Fat	Other nutrients
Cereal	C1			0	12
	C2			3	10
Millet	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereal	P1	66			10
	P2		14		8

The following additional facts are known.

1. Both the pseudo-cereals had higher amounts of carbohydrate as well as higher amounts of protein than any millet.
2. Both the cereals had higher amounts of carbohydrate than any pseudo-cereal.
3. All the missing values of carbohydrate amounts (in grams) for all the food grains are non-zero multiples of 5.
4. All the missing values of protein, fat and other nutrients amounts (in grams) for all the food grains are non-zero multiples of 4.
5. P1 contained double the amount of protein that M3 contains.

Question 66

How many grams of other nutrients were there in 100 grams of nutrients in M3?

The table given below shows the amount, in grams, of carbohydrate, protein, fat and all other nutrients, per 100 grams of nutrients in seven food grains. The first column shows the food grain category and the second column its codename. The table has some missing values.

Food grain category	Codename	Carbohydrate (g)	Protein (g)	Fat (g)	Other nutrients (g)
Cereal	C1	66	14	1	1
Pseudo-cereal	P1	66	14	1	1
Millet	M1	66	14	1	1
Pseudo-cereal	P2	66	14	1	1
Cereal	C2	66	14	1	1
Pseudo-cereal	P3	66	14	1	1
Millet	M2	66	14	1	1

The following additional facts are known.

- Both the pseudo-cereals had higher amounts of carbohydrate as well as higher amounts of protein than any millet.
- Both the cereals had higher amounts of carbohydrate than any pseudo-cereal.
- All the missing values of carbohydrate amounts (in grams) for all the food grains are non-zero multiples of 5.
- All the missing values of protein, fat and other nutrients amounts (in grams) for all the food grains are non-zero multiples of 4.
- P1 contained double the amount of protein that M2 contains.

VIDEO SOLUTION

Subscribe to MBA Exams Youtube Channel

Instructions

There are only three female students - Amala, Koli and Rini - and only three male students - Biman, Mathew and Shyamal - in a course. The course has two evaluation components, a project and a test. The aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

The projects are done in groups of two, with each group consisting of a female and a male student. Both the group members obtain the same score in the project.

The following additional facts are known about the scores in the project and the test.

- The minimum, maximum and the average of both project and test scores were identical - 40, 80 and 60, respectively.
- The test scores of the students were all multiples of 10; four of them were distinct and the remaining two were equal to the average test scores.
- Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Yet Amala had the highest aggregate score.
- Shyamal scored the second highest in the test. He scored two more than Koli, but two less than Amala in the aggregate.
- Biman scored the second lowest in the test and the lowest in the aggregate.
- Mathew scored more than Rini in the project, but less than her in the test.

Question 67

What was the weight of the test component?

- A 0.60
- B 0.50
- C 0.75
- D 0.40

↓

	Project	Test	Aggregate
Amala			
Koli			
Rini			
Biman			
Mathew			
Shyamal			

There are only three female students - Amala, Koli and Rini - and only three male students - Biman, Mathew and Shyamal - in a course. The course has two evaluation components, a project and a test. The aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

The projects are done in groups of two, with each group consisting of a female and a male student. Both the group members obtain the same score in the project.

The following additional facts are known about the scores in the project and the test.

- The minimum, maximum and the average of both project and test scores were identical - 40, 80 and 60, respectively.
- The test scores of the students were all multiples of 10; four of them were distinct and the remaining two were equal to the average test scores.
- Amala's score in the project is double that of Koli in the same, but Koli scored 20 less than Amala in the test. Yash scored the second highest in the test. He scored two more than Koli, but less than Amala in the aggregate.
- Mathew scored the second highest in the test and the lowest in the aggregate.
- Mathew scored more than Biman in the project, but less than her in the test.

VIDEO SOLUTION

Instructions

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of these people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran		1	
Dhinesh			2
Ezhil			1
Fani	1		
Gowtham		3	2

The following additional facts are also known.

- Amudha, Bharatan, and Chandran are mutual strangers.
- Amudha, Dhinesh, and Fani are Ezhil's friends.
- Chandran and Gowtham are friends.
- Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
- Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

Question 68

Who is an acquaintance of Amudha?

- A Dhinesh
- B Fani
- C Gowtham
- D Ezhil

	A	B	C	D	E	F	G
A							
B							
C							
D							
E							
F							
G							

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of the people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran	1		2
Dhinesh			1
Ezhil			2
Fani	3		
Gowtham		2	

Following additional facts are also known.

1. Amudha, Bharatan, and Chandran are mutual strangers.

2. Amudha, Dhinesh, and Fani are Ezhil's friends.

3. Chandran and Gowtham are friends.

4. Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.

5. Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

VIDEO SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.

The Dean's office recently scanned student results into the central computer system. When their character reading software cannot read something, it leaves the space blank. The scanner output read as follows:

Name	Finance	Marketing	Statistics	Strategy	Operations	GPA
Aparna		B	F			1.4
Bikas	D	D	F	F		
Chandra		D	A	F	F	2.4
Deepak	A	B		D	D	3.2
Fazal	D	F		B	D	2.4
Gowri	C	C	A		B	3.8
Hari		B	A		D	2.8
Ismet			B		A	
Jagdeep	A	A	B		C	3.8
Kunal	F		A	F	F	1.8
Leena	B	A		B	F	3.2
Manab			A	B	B	
Nisha	A	D	B	A	F	3.6
Osman	C		B	B	A	4.6
Preeti	F	D		D		3.2
Rahul	A	C	A		F	4.2
Sameer		C	F	B		
Tara	B					2.4
Utkarsh			F	C	A	3.0
Vipul	A		C	C	F	2.4

In the grading system, A, B, C, D, and F grades fetch 6, 4, 3, 2, and 0 grade points respectively. The Grade Point Average (GPA) is the arithmetic mean of the grade points obtained in the five subjects. For example Nisha's GPA is $(6 + 2 + 4 + 6 + 0) / 5 = 3.6$. Some additional facts are also known about the students' grades. These are

- Vipul obtained the same grade in Marketing as Aparna obtained in Finance and Strategy.
- Fazal obtained the same grade in Strategy as Utkarsh did in Marketing.
- Tara received the same grade in exactly three courses.

Question 69

In operations, Tara could have received the same grade as

- Ismet
- Hari
- Jagdeep
- Manab

Directions for the following four questions: Answer the questions on the basis of the information given below.

The Dean's office recently scanned student results into the central computer system. When their character reading software cannot read something, it leaves the space blank. The scanner output read as follows:

Name	Subject	Grade	Score	Percentage
1. Nisha	Maths	A	10	100
2. Vikram	Science	B	8	80
3. Anjali	History	C	7	70
4. Rohan	Art	D	6	60
5. Priya	Music	E	5	50
6. Arjun	Physical Education	F	4	40
7. Neha	Computer	A	9	90
8. Adarsh	Maths	B	8	80
9. Isha	Science	C	7	70
10. Karan	History	D	6	60
11. Pooja	Art	E	5	50
12. Ravi	Music	F	4	40
13. Sanya	Physical Education	A	10	100
14. Abhinav	Computer	B	8	80
15. Divya	Maths	C	7	70
16. Harsh	Science	D	6	60
17. Khushi	History	E	5	50
18. Naman	Art	F	4	40
19. Nisha	Maths	A	10	100
20. Vikram	Science	B	8	80
21. Anjali	History	C	7	70
22. Rohan	Art	D	6	60
23. Priya	Music	E	5	50
24. Arjun	Physical Education	F	4	40

In the grading system, A, B, C, D, and F grades fetch 6, 4, 3, 2, and 0 grade points respectively. The Grade Point Average (GPA) is the arithmetic mean of the grade points obtained in the five subjects. For example Nisha's GPA is $(6 + 2 + 4 + 6 + 0) / 5 = 3.6$. Some additional facts are also known about the students' grades. These are:

- Vipul obtained the same grade in Marketing as Aparna obtained in Finance and Strategy.

Handwritten notes on the right side of the screen:

A → 6
B → 4
C → 3
D → 2
F → 0

VIDEO SOLUTION

CAT 2025 MOCKS & SECTIONALS
WITH VIDEO SOLUTIONS & ANALYSIS

₹ 2999

ENROLL NOW

Instructions

In a coaching class, some students register online, and some others register offline. No student registers both online and offline; hence the total registration number is the sum of online and offline registrations. The following facts and table pertain to these registration numbers for the five months - January to May of 2023. The table shows the minimum, maximum, median registration numbers of these five months, separately for online, offline and total number of registrations. The following additional facts are known.

- In every month, both online and offline registration numbers were multiples of 10.
- In January, the number of offline registrations was twice that of online registrations.
- In April, the number of online registrations was twice that of offline registrations.
- The number of online registrations in March was the same as the number of offline registrations in February.
- The number of online registrations was the largest in May.

	Minimum	Maximum	Median
Online	40	100	80
Offline	30	80	50
Total	110	130	120

Question 70

What was the total number of registrations in April?

shows the minimum, maximum, median registration numbers of these five months, separately for online, offline and total number of registrations. The following additional facts are known.

- In every month, both online and offline registration numbers were multiples of 10.
- In January, the number of offline registrations was twice that of online registrations.
- In April, the number of online registrations was twice that of offline registrations.
- The number of online registrations in March was the same as the number of offline registrations in February.
- The number of online registrations was the largest in May.

	Minimum	Maximum	Median
Online	40	100	80
Offline	30	80	50
Total	110	130	120

Handwritten notes on the right side of the screen:

Online: 40, 80, 100
Offline: 30, 50, 80
Total: 110, 130, 120

Question 71

Which pair of months definitely had the same total number of registrations?

- I. January and April
II. February and May

- A Both I and II
B Only II
C Only I
D Neither I nor II

shows the minimum, maximum, median registration numbers of these five months, separately for online, offline and total number of registrations. The following additional facts are known.

- In every month, both online and offline registration numbers were multiples of 10.
- In January, the number of offline registrations was twice that of online registrations.
- In April, the number of online registrations was twice that of offline registrations.
- The number of online registrations in March was the same as the number of offline registrations in February.
- The number of online registrations was the largest in May.

	Minimum	Maximum	Median
Online	30	100	80
Offline	40	80	50
Total	110	130	120

Handwritten notes on the right side of the video frame show a table with columns: Online, Offline, Total. The values are: Online: 40, 80, 100; Offline: 30, 50, 80; Total: 70, 130, 180. A large play button is overlaid on the video frame.

Question 72

What was the number of online registrations in January?

shows the minimum, maximum, median registration numbers of these five months, separately for online, offline and total number of registrations. The following additional facts are known.

- In every month, both online and offline registration numbers were multiples of 10.
- In January, the number of offline registrations was twice that of online registrations.
- In April, the number of online registrations was twice that of offline registrations.
- The number of online registrations in March was the same as the number of offline registrations in February.
- The number of online registrations was the largest in May.

	Minimum	Maximum	Median
Online	30	100	80
Offline	40	80	50
Total	110	130	120

Handwritten notes on the right side of the video frame show a table with columns: Online, Offline, Total. The values are: Online: 40, 80, 100; Offline: 30, 50, 80; Total: 70, 130, 180. A large play button is overlaid on the video frame.

CAT MOCKS & SECTIONALS
WITH VIDEO SOLUTIONS & ANALYSIS

₹ 1499

ENROLL NOW

Instructions

There are only three female students - Amala, Koli and Rini - and only three male students - Biman, Mathew and Shyamal - in a course. The course has two evaluation components, a project and a test. The aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

The projects are done in groups of two, with each group consisting of a female and a male student. Both the group members obtain the same score in the project.

The following additional facts are known about the scores in the project and the test.

1. The minimum, maximum and the average of both project and test scores were identical - 40, 80 and 60, respectively.
2. The test scores of the students were all multiples of 10; four of them were distinct and the remaining two were equal to the average test scores.
3. Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Yet Amala had the highest aggregate score.
4. Shyamal scored the second highest in the test. He scored two more than Koli, but two less than Amala in the aggregate.
5. Biman scored the second lowest in the test and the lowest in the aggregate.
6. Mathew scored more than Rini in the project, but less than her in the test.

Question 73

What was the maximum aggregate score obtained by the students?

- A 68
- B 80
- C 62
- D 66

↓

	Project	Test	Aggregate
Amala			
Koli			
Rini			
Biman			
Mathew			
Shyamal			

cracku

There are only three female students - Amala, Koli and Rini - and only three male students - Biman, Mathew and Shyamal - in a course. The course has two evaluation components, a project and a test. The aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

The projects are done in groups of two, with each group consisting of a female and a male student. Both the group members obtain the same score in the project.

The following additional facts are known about the scores in the project and the test.

1. The minimum, maximum and the average of both project and test scores were identical - 40, 80 and 60, respectively.
2. The test scores of the students were all multiples of 10; four of them were distinct and the remaining two were equal to the average test scores.
3. Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Yet Amala had the highest aggregate score.
4. Shyamal scored the second highest in the test. He scored two more than Koli, but two less than Amala in the aggregate.
5. Biman scored the second lowest in the test and the lowest in the aggregate.
6. Mathew scored more than Rini in the project, but less than her in the test.

VIDEO SOLUTION

Question 74

What was Mathew's score in the test?

↓

	Project	Test	Aggregate
Amala			
Koli			
Rini			
Biman			
Mathew			
Shyamal			

There are only three female students - Amala, Koli and Rini - and only three male students - Biman, Mathew and Shyamal - in a course. The course has two evaluation components, a project and a test. The aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

The projects are done in groups of two, with each group consisting of a female and a male student. Both the group members obtain the same score in the project.

The following additional facts are known about the scores in the project and the test.

1. The minimum, maximum and the average of both project and test scores were identical - 40, 80 and 60, respectively.

2. The test scores of the students were all multiples of 10; four of them were distinct and the remaining two were equal to the average test scores.

3. In the project, the score of Amala was double that of Koli in the same, but Koli scored 20 more than Mathew in the project.

4. In the test, Y scored the second highest in the test. He scored two more than Koli, but less than Amala in the aggregate.

5. Amala scored the second highest in the test and the lowest in the aggregate.

6. Mathew scored more than Biman in the project, but less than her in the test.

VIDEO SOLUTION

Question 75

Which of the following pairs of students were part of the same project team?

- i) Amala and Biman
- ii) Koli and Mathew

- A Only ii)
- B Only i)
- C Neither i) nor ii)
- D Both i) and ii)

↓

	Project	Test	Aggregate
Amala			
Koli			
Rini			
Biman			
Mathew			
Shyamal			

There are only three female students - Amala, Koli and Rini - and only three male students - Biman, Mathew and Shyamal - in a course. The course has two evaluation components, a project and a test. The aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

The projects are done in groups of two, with each group consisting of a female and a male student. Both the group members obtain the same score in the project.

The following additional facts are known about the scores in the project and the test.

1. The minimum, maximum and the average of both project and test scores were identical - 40, 80 and 60, respectively.

2. The test scores of the students were all multiples of 10; four of them were distinct and the remaining two were equal to the average test scores.

3. In the project, the score of Amala was double that of Koli in the same, but Koli scored 20 more than Mathew in the project.

4. In the test, Y scored the second highest in the test. He scored two more than Koli, but less than Amala in the aggregate.

5. Amala scored the second highest in the test and the lowest in the aggregate.

6. Mathew scored more than Biman in the project, but less than her in the test.

VIDEO SOLUTION

CAT 2025 MOCKS & SECTIONALS
WITH VIDEO SOLUTIONS & ANALYSIS

₹ 2999

ENROLL NOW